

Predicting Zestimate Price Factors Using Zillow and Census Data



DAT 490 Capstone

Team JKJK

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Abstract and Executive Summary

Zillow Group, Inc., founded in 2006, is the leading digital real estate marketplace in the United States. It also provides home value estimates through its Zestimate tool, an automated home valuation model, serving millions of users through platforms like Zillow, Trulia and StreetEasy. This study uses a Zillow dataset from HuggingFace.co in conjunction with U.S. Census data, to analyze the socioeconomic and regional factors influencing Zestimate home value prices and rental price dynamics across the United States. The dataset has 7 subsets of data and over 2 million rows of data. The project centers around two key research questions: (1) How do poverty rates and population density influence Zillow home value estimates across different regions? And (2) How did the COVID-19 pandemic and its related socio-economic factors impact rental price trends across cities and counties? The data for research question one focused on the average ZHVI by city (nationwide) from 2018 to 2023. ZHVI indicates the typical home value over a region, in this case city and county.

Employing linear regression, random forest modeling, and clustering techniques such as PCA and K-Means, our analysis explores the explicit and latent relationships between demographic variables and real estate trends, with a focus on how Zestimate responds to these factors. Our best performing model at the city level was a histogram gradient boosting regressor, which explains 34.96% of variance, and had a root mean squared error of .5074. While at the county level, the best performer was a tuned random forest model scoring an RMSE of just .3298 and explaining nearly 72% of variance. At a county level, our analysis showed that poverty rate is the most influential predictor of home and rental values whereas at a city level, population density plays a larger role in determining prices.

Our initial hypothesis for our second research question was that we would see more growth in more relaxed COVID lockdown areas but this was not the case. What we found was more surprising given all of the hype of the rapid growth in the major metro areas in the

southern and midwest regions. We gathered data from various outlets relating to unemployment, housing, population and even COVID lockdown stringency to key in on what thought may have contributed to the rapid rent growth in different areas. This led us to finding that post pandemic shifts, including mass migrations from urban and rural into suburban areas loosely correlated with spikes in rental price volatility and growth. The metrics that we found that mattered for each was and found that the greatest factors for rent growth was population density, and how strict COVID lockdown measures were, surprisingly. The greatest detractors for rent growth in the post pandemic era were housing unity density unemployment trends and unemployment average.

This analysis highlights the importance of incorporating socioeconomic context into residential valuation tools like the Zestimate. While Zillow's current models are a good starting point, transparency and accuracy can be significantly improved with the inclusion of more comprehensive community metrics. We also emphasize the ethical considerations of predictive modeling for housing, especially in historically underserved communities where algorithmic bias can worsen existing inequalities. These findings not only yield strategic value to Zillow's ongoing model development, but also inform more nuanced insight into how real estate markets respond to social and economic shocks.

Project Plan

Company Details:

Founded: February 2006

Founders: Rich Barton, Lloyd Frink, Spencer Rascoff

Headquarters: Seattle, WA

IPO: Public as of July 20, 2011

Categories: Real Estate, Home Valuation

Address:

Zillow, Inc.

1301 Second Avenue

Floor 36

Seattle, WA 98101

United States

Company Communication:

Phone Number: 877-313-8601 or 206-470-7000

Website: <http://zillow.com/>

Business Description:

Zillow Group, Inc. (Zillow) is a digital real estate marketplace that connects people with information to buy, sell, rent, and finance homes, condos, and other properties. One of the most influential offerings is the *Zestimate*®, an automated home valuation model that displays property estimates using publicly available data, user information, and property algorithms. Zillow, along with its subsidiaries Trulia and StreetEasy, serves millions monthly. These services support more than home buyers and renters, but also real estate professionals through advertising and lead generation tools.

Financials¹:

Ticker Symbol: ZG

Latest Financial Data: March 2025

Revenue: \$598M (+13.04%)

Assets: \$5.74B (-1.42%)

Liabilities: \$992M (+1.11%)

Return on Equity: (+0.17%)

Earnings: (+.03)/share

No. of Employees: 6,856

Key Executives:

[Jenny Arden](#), Chief Design Officer

[David Beitel](#), Chief Technology Officer

[Jun Choo](#), Chief Operating Officer

[Jeremy Hofmann](#), Chief Financial Officer

[Brad Owens](#), General Counsel

[Jennifer Rock](#), Chief Accounting Officer

[Errol Samuelson](#), Chief Industry Development Officer

[Dan Spaulding](#), Chief People Officer

[Jeremy Wacksman](#), Chief Executive Officer

Major Competitors:

Redfin

[Realtor.com](#)

Trulia (technically a subsidiary, still a competitor)

Opendoor

Rocket Homes

Compass

Homesnap

Business/Analysis Opportunity:

Utilizing a dataset published on Hugging Face and sourced from Zillow, our team is examining property-level Zestimate values across the U.S. The dataset includes housing estimates, property details, and geographic information, which we will enrich with ancillary data from [Census.gov](https://www.census.gov).

We aim to address Zillow's need for deeper insights into how housing markets across the country are affected by local and regional factors and how they impact the accuracy of a *Zestimate*. By developing predictive models and comparing regional data, we aim to support Zillow in offering a more transparent and personalized valuation.

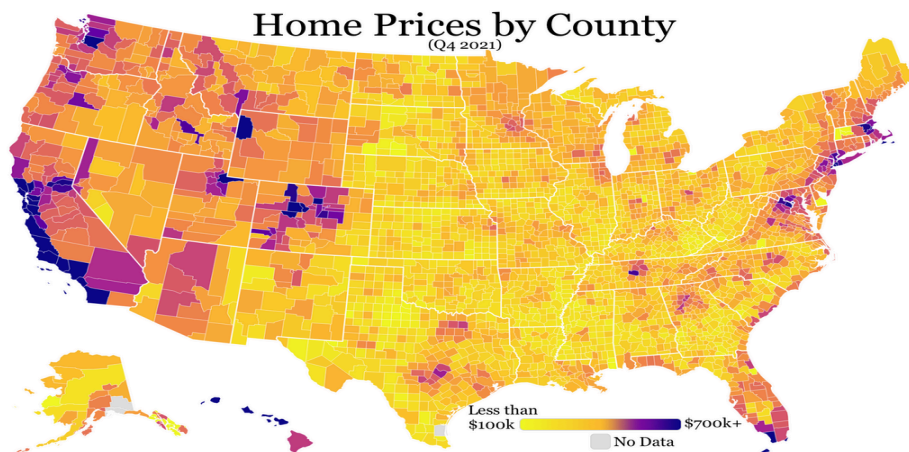
Research Questions

The United States real estate market has many factors that influence buyer and seller behavior. All of the consumers are affected by the macro-level economics while each individual market has micro-level factors that affect that particular market. Our research seeks to employ given datasets to pinpoint contributing factors in the real estate market, assess their influence on consumer behavior, and reveal nuanced relationships that shape buyer and seller interactions.

RQ1: How do poverty rates and population density influence Zillow home value estimates across different regions?

As Zillow's home price estimate tool continues to evolve, it is increasingly important to understand the influence of external factors to maintain public trust and enhance the tool's accuracy. The value of a home is rarely based solely on property features but rather through an aggregation of social and economic context of the surrounding area. Supply limitations in a dense urban area will increase valuation, yet high poverty levels in a nearby neighborhood could suppress those same pricing pressures.

This research question is aimed at revealing the relationship between key socioeconomic indicators and their effect on ZHVI (*Zestimate*) of homes across various geographic regions. Using the pattern recognition, this analysis should allow Zillow to adjust or better explain *Zestimates* to end users based on broader community characteristics, thereby enhancing the tool for buyers, sellers, and investors who use the platform.



2. https://en.wikipedia.org/wiki/List_of_U.S._states_by_median_home_price#/media/File:Home_prices_by_county.webp

RQ2: To what extent did the COVID-19 pandemic and its associated socio-economic shifts (including changes in migration patterns) lead to a significant and quantifiable change in average rental prices and/or rental price growth rates in major U.S. metropolitan areas, when comparing the pre-pandemic period (2018-2019) to the post-shutdown period (2020-2022)?

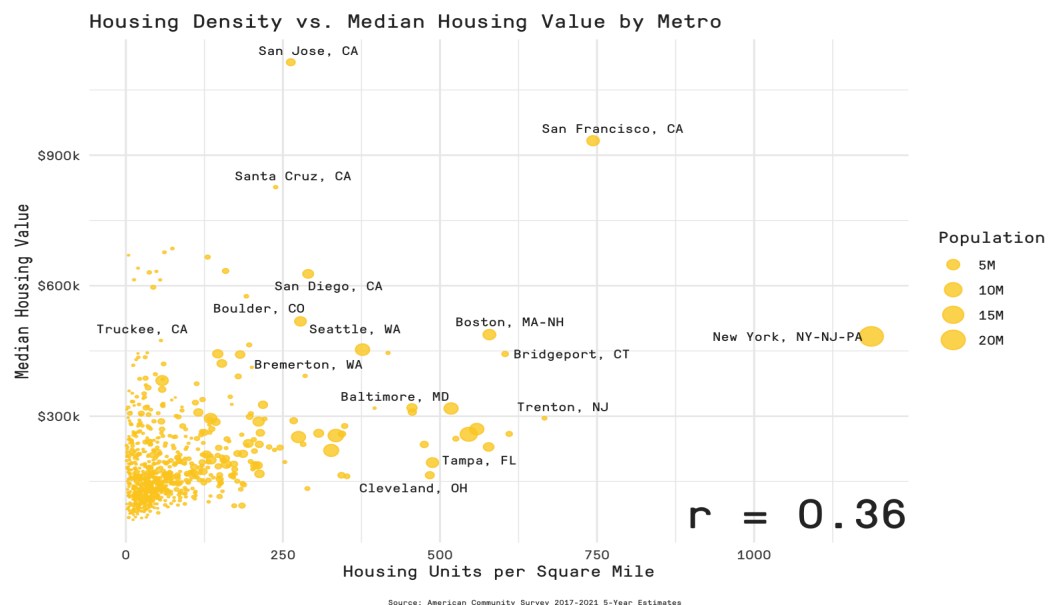
Cities in the United States go through changes over time. Some changes seem to happen very slowly while others seem to happen very quickly. In our research question 2, we aim to identify any trends that were happening prior to the COVID pandemic and aim to analyze any trends that sped up or reversed during the years post COVID (2020). In doing so we will be able to see which cities/regions are growing more rapidly and what factors contribute to the boom/bust cycle.

Hypotheses

H1: Zillow home values will be positively correlated with population density and negatively correlated with poverty rate. While urban regions will have a higher median Zestimate values than rural regions, with controlled property size

Dense, urban areas are generally marked with low supply and high demand. Contrarily, regions with high poverty likely have lower home valuation due to decreased demand and property investment. Through regional analysis, we seek to test whether our expectations are consistent throughout the data.

In addition to supply and demand price pressures, dense regions are also subject to housing premiums as a result of amenity access, job proximity, and transportation infrastructure. We will test whether *Zestimates* are consistently higher in urban regions vis rural, even when home size, features, and land are accounted for. This will assist in determining if urban location is a strong predictor of value.



7. Housing Density vs. Housing Value by Metros: <https://www.kaseyklimes.com/notes/2023/5/2/density-doesnt-drive-housing-prices>

H2: States that had more relaxed lockdown policies experienced greater population migration and thus in turn increase in rental prices.

The feeling of economic or social restriction experienced by COVID 19 lockdown policies drove people away from states with stricter lockdowns into places with more relaxed policies, creating a significant impact on the rental market. These trends would be somewhat slowed down or reversed following the end of COVID lockdown policies. Thus, particularly areas with the largest urban boom during covid may have also experienced the greatest decline in housing market value.

Data

The data used in this project is sourced from Zillow's publicly available housing dataset, specifically, one posted by misikoff on Hugging Face³, and includes many detailed housing metrics. It is divided into multiple parquets, each tracking different aspects of the U.S. real estate market; sale prices, listings, rental prices, days on market, and new construction listings. These will allow for the analysis of home valuation and renting behavior across geographic and socioeconomic variations, which aligns with our research questions.

Home Values

The *home_values.parquet* provides Zillow's estimated home values (ZHVI) for properties spanning different tiers; bottom, middle and top. These values are smoothed and seasonally adjusted. Each row provides a property type, number of bedrooms, state, region type, region ID, and a size rank, allowing for analysis of value trends over time across the various housing segments.

Sales

The *sales.parquet* data includes historical values such as median sale price, sale-to-list ratios, and percentage sold above and below list price (raw and smoothed). This will be used to cross-reference with the home values and validate the pricing trends and market throughout the housing market.

For Sale Listings

For_sale_listings.parquet tracks the new houses added to inventory and includes variables for new listings, median listing price, and new pending sales. This will help gauge the supply of a regional market and will assist in comparing listing activity between rural and urban areas.

Days on Market

In order to further understand the supply and demand dynamics, *days_on_market.parquet* tracks how long a property was listed before pending a sale. In addition, the included price cut data will increase the visibility of over-supply of homes, if applicable.

Rentals

This parquet, *rentals.parquet*, tracks the rental market. The variables included are region ID, region, region type, size rank, property type, state, home type, date, and rent (smoothed and seasonally adjusted) . This dataset contains 1.26 million rows of data, which encompasses the years 2015 to 2023. This set of data will help to show how migration affected rental prices across different markets.

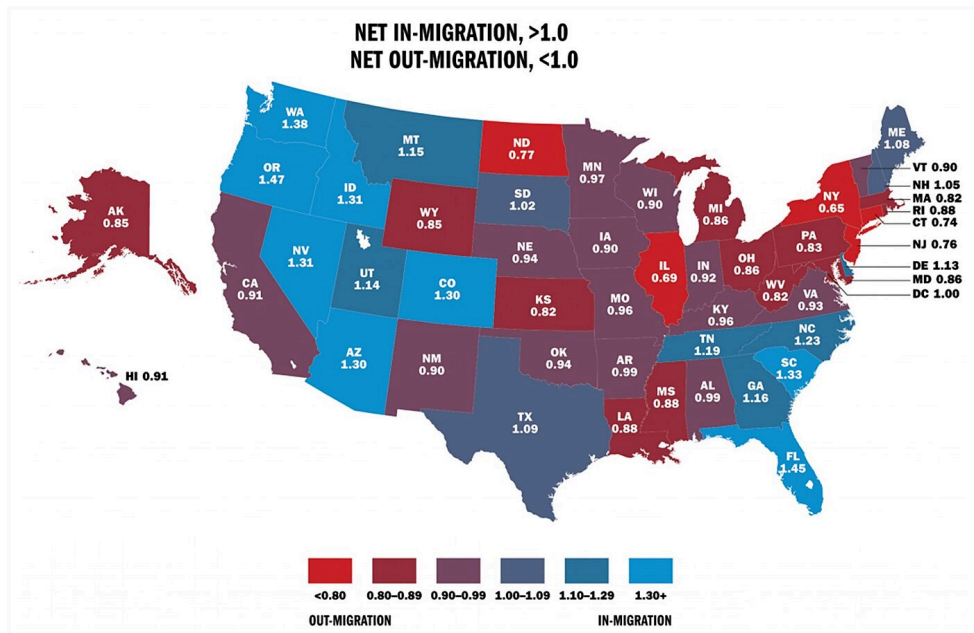
New Construction

The *new_construction.parquet* adds information about new construction properties, including date, median sale price (also adjusted per square foot), and the sales for a particular month in the listed region. This information will be invaluable in solidifying the regional supply and demand context necessary for identifying market trends.

Measurements

For this project we aim to understand how each piece of information contributes to the factors that affect home values and migration patterns. For our first research question we aim to determine how socio-economic factors and population density affect home values in different regions of the United States. Zillow has their proprietary Zillow Home Value Index (ZHVI) from their website “A measure of the typical home value and market changes across a given region and housing type. It reflects the typical value for homes in the 35th to 65th percentile range. Available as a smoothed, seasonally adjusted measure and as a raw measure.”⁴

Our research project aims to also identify key factors that affected migration patterns that either were accelerated or reversed from pre pandemic era to post pandemic era. In doing so we will be measuring net migration, new construction, rental price fluctuations, and median income



5. <https://www.cato.org/blog/mapping-interstate-migration>

Methodology

For our first research question (RQ1) we are trying to determine how poverty rate and population density are linked to home values. In order to determine this we are going to do a regression analysis. We will start with a simple linear regression analysis and also use a random forest analysis to determine any non-linear relationships.

For our second research question (RQ2) we aim to decipher what factors affected migratory patterns in the United States pre and post pandemic. In order to do this we will use

regression analysis to determine the factors that led to migration and we will follow this up with a cluster analysis to determine which regions/metro areas were similar in migration patterns.

In order to analyze these factors we will use structured data from the zillow datasets. We will also use ancillary data from the [census.gov](https://www.census.gov) website. We anticipate that we will merge datasets together to have a more enriched dataset to determine specific patterns. Our exploratory data analysis, we suspect, will give us a better understanding of what factors are correlated. This will give us a better understanding of what variables we should focus on.

Computational Methods and Outputs

For our regression analysis (linear and random forest) in research questions 1 and 2 we will use the R-squared (r^2) and Root Mean Squared Error (RMSE) to determine how well the linear regression performs. The r^2 will help to determine how much of the dependent variable is predicted by the independent variables. The Root Mean Squared Error will help to determine how well our model does in regards to our variables chosen (closer to 1 is better).

In our clustering analysis we will use a visual representation to determine which regions/metros are most similar to each other. We will also use a silhouette score to help to determine how similar the data points are to each other.

Our outputs for research question 1 will be a table of which regions/metros are most correlated with population density/poverty rates to home values in each. We will have a map showing each with corresponding shaded bubbles plotting these factors.

EXHIBIT 6a

Population growth and declines, 3-year and 10-year comparisons

Myrtle Beach, SC, Midland, TX and Greeley, CO grew the fastest in the short and long run.



6. Freddie Mac population growth chart (2010-2019): https://www.freddiemac.com/research/insight/20210128-population-growing?utm_source=chatgpt.com

Campaign Implementation

The U.S. real estate market is highly complex and shaped by a wide range of influencing factors. Each local market operates with its own microeconomic conditions (income levels, housing supply, and population density) while broader macroeconomic trends (inflation, interest rates, and migration patterns) impact the country as a whole. Zillow is uniquely positioned to analyze both levels of data, allowing it to identify national trends while also uncovering the local drivers that influence specific housing markets. This dual insight gives Zillow a competitive

advantage in tailoring strategies, tools, and services to fit both broad market shifts and hyper-local needs.

Based on the findings from our research questions, Zillow can implement targeted strategies to boost engagement, revenue, and market share. For Research Question 1, our analysis of poverty rates and population density will reveal which cities are likely to experience higher-than-average home values and what factors contribute. Zillow can leverage this insight by increasing marketing efforts toward real estate agents and property managers in those high-growth areas. Additionally, by identifying more valuable markets, Zillow can strategically increase its mortgage lending ad spend to attract clients with higher purchasing power, maximizing ROI on lending services.

For Research Question 2, our examination of migration patterns and rental market shifts post-COVID enables Zillow to pinpoint key metro areas experiencing significant renter inflows or pricing volatility. With this knowledge, Zillow can refine its regional advertising and outreach efforts, encouraging property managers and landlords in high-demand areas to list on the platform. By aligning marketing spend with migratory and economic trends, Zillow can effectively position itself as the go-to platform for capturing increased rental activity and growing its footprint in strategic metros.

Literature Review

Since the onset of modern economic growth, the issue that persists on a global scale is the variation of housing prices. Migration towards urban centers, usually driven by larger and more abundant infrastructure have contributed to fluctuations of demand and supply in housing. These dynamics became particularly evident in early 2020, when the COVID-19 pandemic began to affect individuals directly. As a result, the concerns surrounding the accessibility and affordability of housing have remained and intensified.

When it comes to the migration patterns during the COVID-19 pandemic, it is evident that the pandemic has substantially impacted in-migration patterns of large urban centers and cities, causing an economic shift. It is mentioned in the article *Internal Migration in Times of COVID-19* by Rowe (2023), that a stream of ‘urban exodus’ occurred throughout the early phases of the pandemic, causing increases of the number of people leaving large urban centers for more rural and suburban areas. It is evident that the percentage of out-migration from urban centers outnumbers the in-migration rate, leading to urban centers becoming increasingly depopulated. Rowe (2023) also mentions residents having to accommodate the demands of remote working, homeschooling caused by lockdown policies. Such strict policies and demands have caused what is called a “donut effect”, meaning residents moving away from urban centers though surrounding it, creating a donut-like shape of migration pattern.

This phenomenon is no exception even in the United States, as similar trends and patterns show. In the research by Ramani & Bloom (2021) from the Department of Economics in Stanford University, the same donut effect is mentioned and explained. Using USPS and Zillow data and having focused on CBDs (Central Business Districts), it was found that the business

and net population have reduced by a significant amount compared to pre-pandemic percentages.

“CBDs of the top 12 US cities have seen net population and business outflows cumulating to around 9% and 16% of their pre-pandemic levels. In contrast, the bottom 50% of zip codes by density have gained about 1-2% of their pre-pandemic stock for population and businesses.” (Ramani & Bloom, 2021)

Knowing this information, for the purposes of this project, we are focusing on the density of the population affected by migration in various areas, predicting the potential fluctuations of housing using the Zillow Housing dataset. Throughout the research, it is crucial to note the principles of population density and how it affects housing prices. It is no secret that regardless of the consideration of the donut effect and migration trends, the population density of larger cities and countries have increased substantially as the economic growth continues.

From the article *Population Density and Housing* (Miles, 2012), a key concept of ‘supply elasticity’ is mentioned. As the population density increases, the more likely that demand for housing increases as well, causing limitations of supplies in real-estate. Naturally, this causes housing prices to increase. This phenomenon can be considered as low supply elasticity, meaning larger cities will have a hard time supplying the demands of consumers, due to land space, cost, and regulations etc. overall resulting in price increases. Generally larger urban cities such as New York and Los Angeles tend to have a relatively lower supply elasticity due to the population size and density, whereas more suburban and rural areas have higher supply elasticity.

Additional issues such as poverty may rise in concentrated areas. It is important to keep in mind that the donut effect of migration has caused substantial amounts of population and businesses to out-migrate from urban centers. There is a good chance that people who experience poverty are included in the out-migration population. Furthermore, poverty is frequently correlated with limited access to necessary services, stable employment - factors that affect the abilities in home ownership, resulting in exposure to vulnerability in housing price fluctuations. The concentration of poverty in certain areas therefore, may experience downward trends in housing prices due to the lack of investment and job opportunities. Part of our research will study the trends of areas where poverty is apparent, estimating the potential impact on housing price estimates.

In conclusion, this literature review analyzes the deep multifaceted relationship between internal migration during the COVID-19 pandemic, population density, poverty, and home values. This study will seek to identify trends and potential constraints that impact fluctuations in home values by connecting this research to the analysis of our Zillow housing data. It is essential to successfully understand the dynamics of this research for strategic urban planning, encouraging economic growth in different areas.

Exploratory Data Analysis

For this project we gathered our dataset from the following dataset repository at Hugging Face <https://huggingface.co/datasets/misikoff/zillow-viewer>. The data consists of structured data collected and compiled by [Zillow.com](https://www.zillow.com). The data was retrieved from the database via .parquet files and explored using Python in Jupyter Notebook. It consists of approximately 2.9 million rows, divided into 7 subsets. Each subset has common “Base Columns” that are columns on each subset. Each subset has its own “Value Columns” relevant to its specific data context. To effectively explore the dataset, our team divided the subsets among members to examine patterns, relationships, and trends. In the sections that follow, we describe the “Base Columns”, define key variables, and highlight our findings from each data subset during the exploratory data analysis process. (Additional Variable names and descriptions found in Appendix 1)

Base Column Variables (with dtypes):

- **Region ID** (*string*): a unique identifier for the region
- **Size Rank** (*int32*): a rank of the regions’s size
- **Date** (*timestamp[ms]*): the date of these forecasts
- **Region** (*string*): Name of the region.
- **Region Type** (*class_label*): the type of region
- **Home type** (*class_label*): The type of home
- **State** (*string*): The US state abbreviation for the state containing the region

At the outset of our project, we developed three potential research questions based on the dataset. After further discussion, we refined our focus to two questions that offered greater clarity and analytical depth:

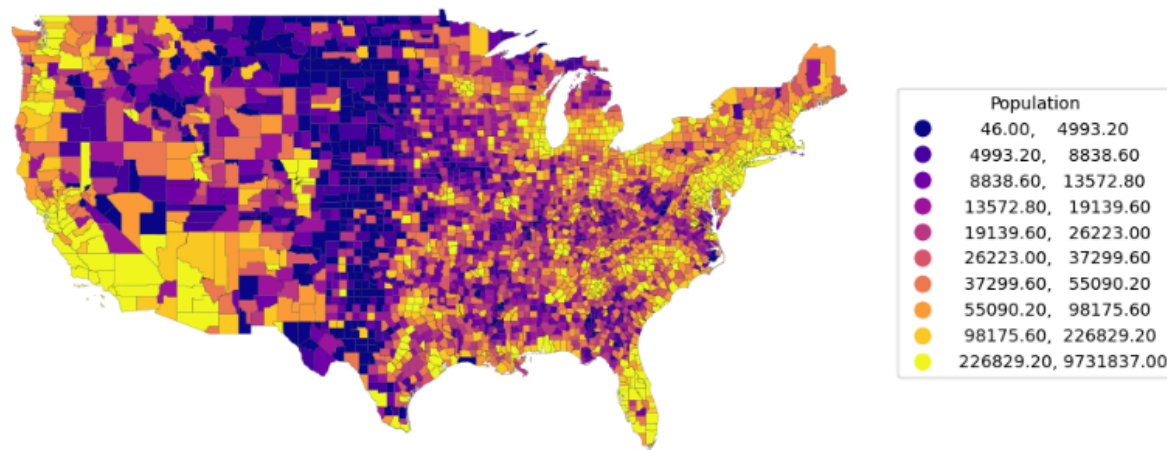
1. *How do poverty rates and population density influence Zillow home value estimates across different regions?*
2. *To what extent did the COVID-19 pandemic and its associated socio-economic shifts—such as migration patterns—contribute to significant changes in average rental prices and rental price growth in major U.S. metropolitan areas between the pre-pandemic (2018–2019) and post-shutdown (2020–2022) periods?*

These questions guided our exploratory analysis and helped us identify patterns and relationships worth investigating further in the modeling phase. In the following sections, we present our findings, first addressing Research Question 1, then moving onto Research Question 2. Lastly we will discuss our Key Findings and Limitations and Conclusion.

Exploratory Data Analysis for Research Question #1

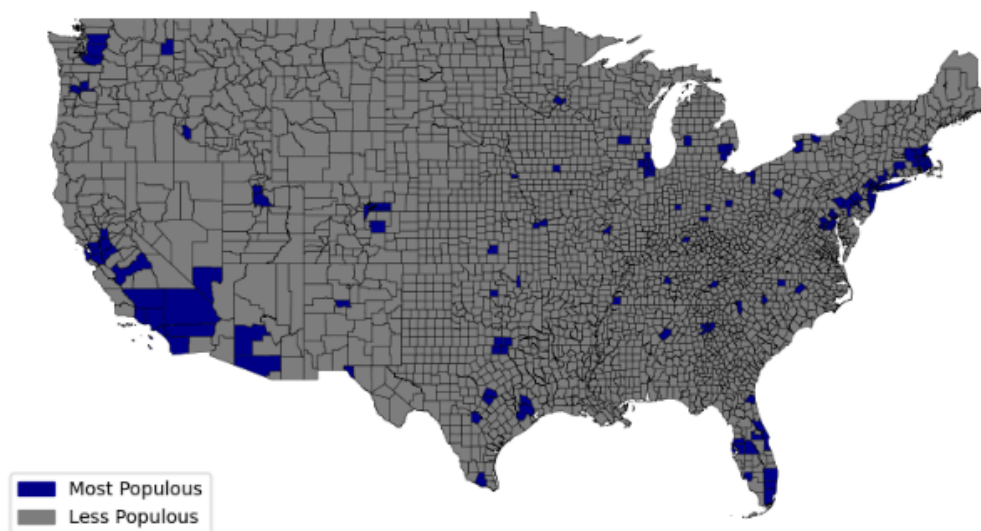
Our first research question “*How do poverty rates and population density influence Zillow home value estimates across different regions?*” leads us to decipher in what ways these factors can influence pricing metrics in different areas and what exactly affects those prices. Initially one of the first things we want to do is see where the population of the continental United States lives. To do this we imported geospatial and population data from the US Census Bureau. Below we found the population of each county and created a heatmap showing how the counties are separated population wise. The majority of the population lives on the coasts. Diving deeper we wanted to see exactly where over 50% of the population lives in the United States by county.

2023 Population Estimates by County



Population of the Continental United States by County

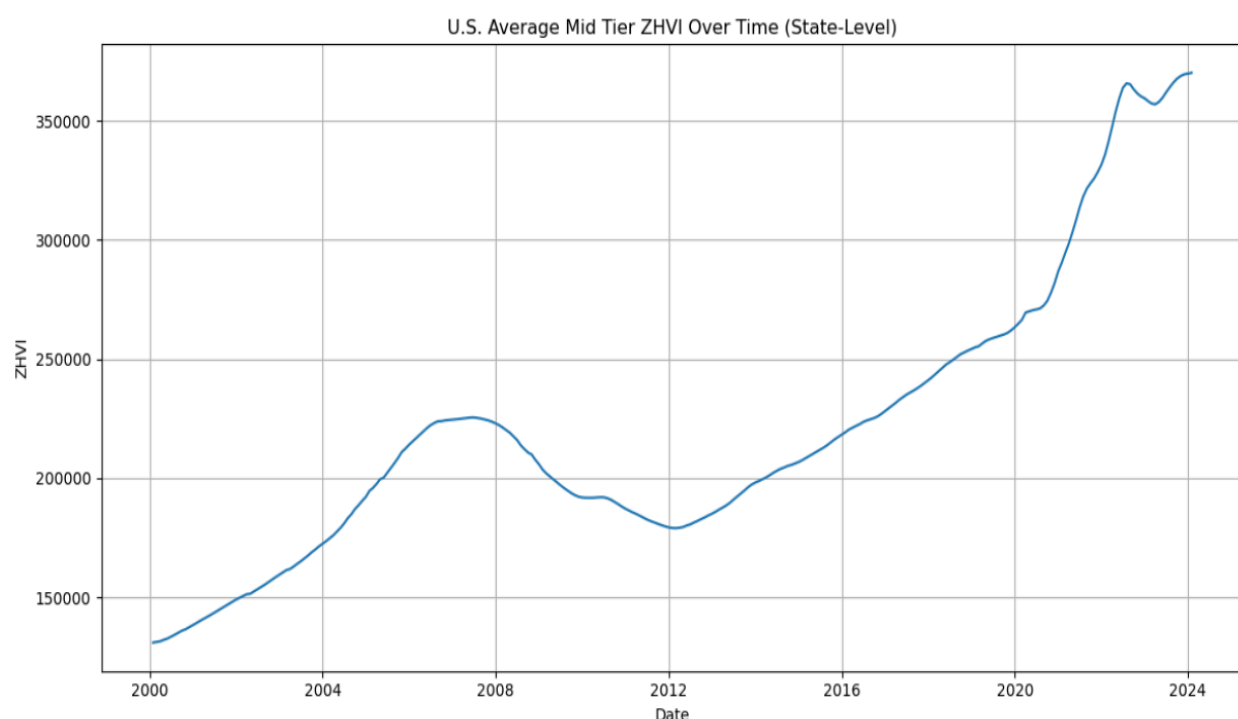
Counties with >50% of CONUS Population



United States Counties with over 50% of the Population

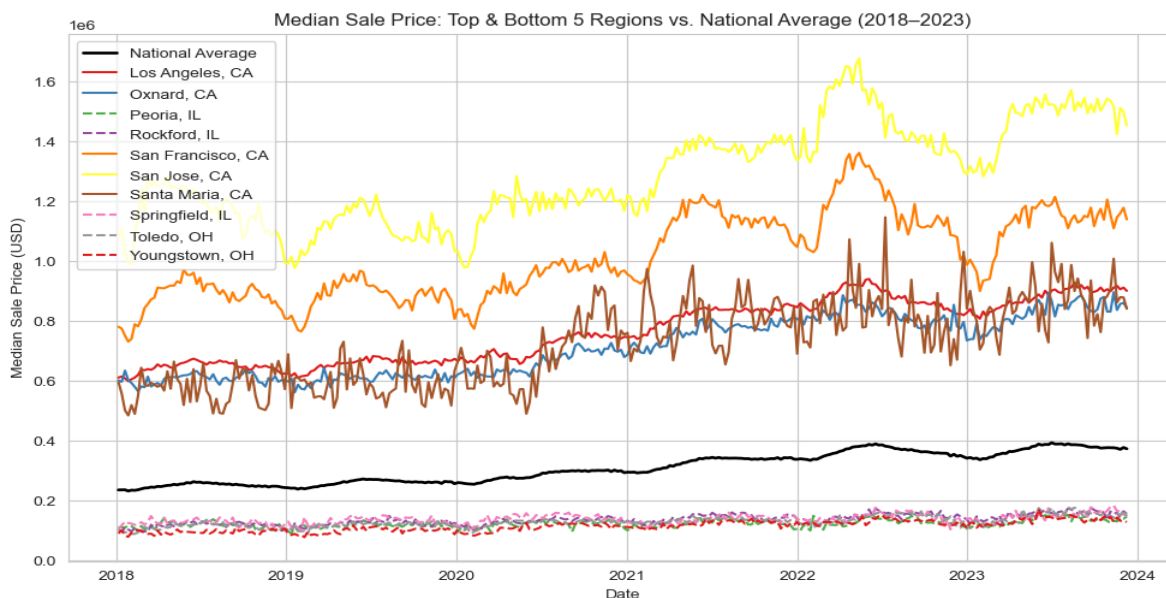
After our analysis here we wanted to explore how home values in different areas of the countries have deviated over time. In order to do this we needed to see the data from different subsets of our data. Utilizing the Home_Values parquet we looked at

different levels of data. To understand long-term housing market trends, we examined Zillow's Home Value Index (ZHVI) for mid-tier homes, smoothed and seasonally adjusted. The chart below shows a steady rise post-2012, with a sharp increase during and after the COVID-19 pandemic.



To gain a better sense of understanding we zoomed in on the period from 2018-2024, see appendix A, we see the steepest price increases beginning mid-2020. This coincides with the post pandemic boom that was seen across the nation.

To highlight regional disparities, we compared the median sale prices for the top five and bottom 5 regions, using the national average as a benchmark. The chart below reveals a widening gap between high-cost and low-cost areas, particularly post-2020. The top markets, primarily located in California, experienced sharp sales price spikes post-2020; whereas low cost areas, primarily in Ohio and Illinois, had more stable sales prices.

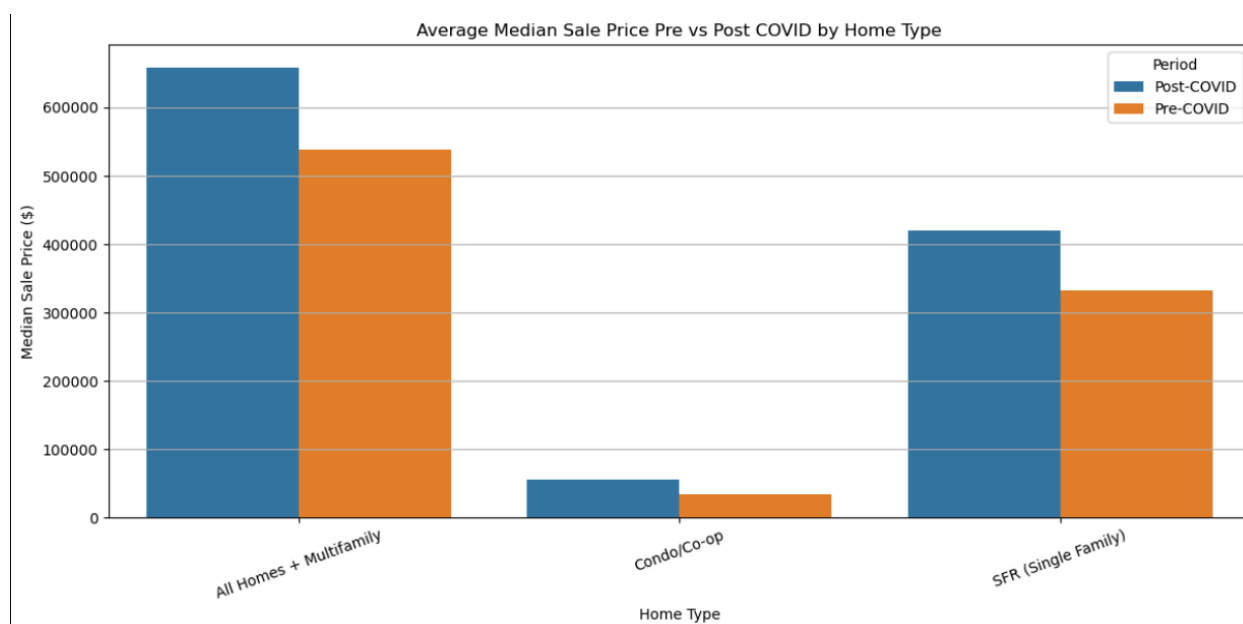


When we talk about home values we, in essence, we're ultimately examining the underlying supply and demand dynamics. To better understand these forces, we analyzed the `days_on_market` dataset, focusing on which states had the longest and shortest home sale timelines (Appendix B). As expected, homes in sparsely populated states tended to stay on the market longer. However, states with the largest populations didn't always have the faster sales cycles. This discrepancy may be influenced by factors such as negotiation periods, inspection delays, appraisal timelines, or increased competition among sellers. Below you will find our chart of the average median days on market.

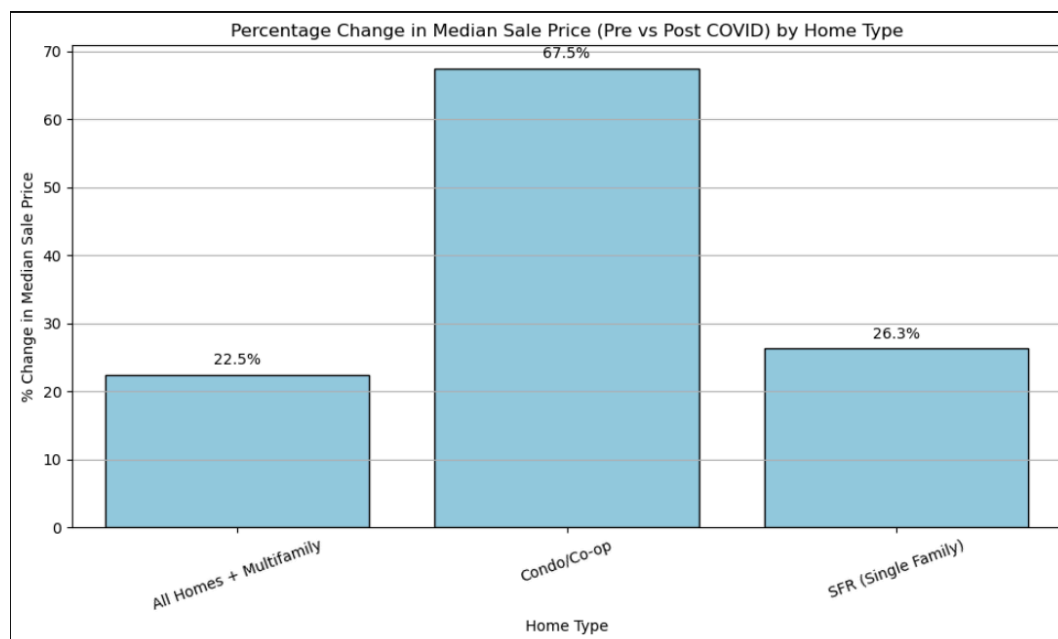
To better understand supply and demand dynamics in a high-growth, high-population state, we analyzed Texas using the `days_on_market` dataset. The chart in Appendix C shows median days pending (blue) alongside listing price cuts (red). Post-2020, we observe a dramatic drop in time to pending during mid-2020 to mid-2022, reaching record lows. This period aligns with post-pandemic heightened demand and rapid market activity. Although the market shows seasonality, the magnitude of change during the pandemic highlights how demand surged even as listing price cuts remained relatively stable. This suggests that in high-density areas like Texas, demand may temporarily override traditional pricing pressures.

When analyzing the average median sale price by home type before and after COVID-19, we observed a significant increase in prices across all categories. The most notable

growth occurred in Single-Family and Multifamily residences. This suggests that people were demanding more spacious and standalone living environments, possibly driven by remote work trends and changing household preferences. In contrast, Condo/Co-Op residences experienced a notable but far smaller price increase. This reflects how buyer preferences may have changed during and after the pandemic, especially in more densely populated urban areas. The difference in growth rates between housing types points to shifting market preference.

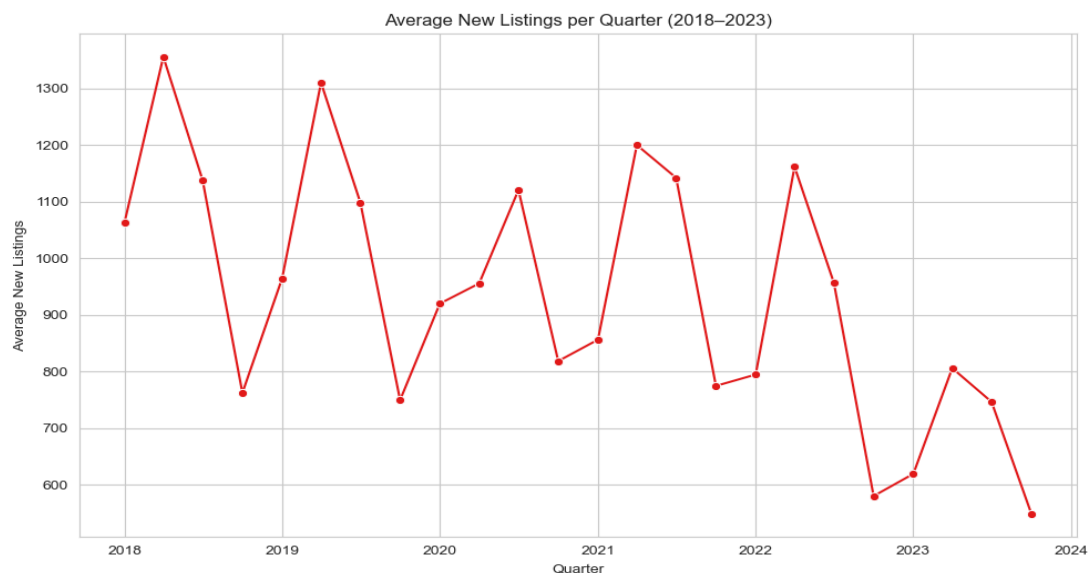


A supplementary bar chart compares the percentage increase in median sale price by home type before and after the COVID-19 pandemic. Interestingly, Condo/Co-Op properties showed the largest percentage growth, despite having the lowest overall prices among the three categories. This suggests that their lower starting price point may have made them more attractive to buyers during the post-pandemic recovery. Additionally, this surge could reflect a rebound effect, as demand for shared living spaces initially declined during the pandemic but later returned as market conditions stabilized and affordability pressures grew in higher-priced segments.



This time-series plot illustrates the average number of new home listings per quarter from 2018 to 2023. Each point represents the national average volume of homes listed during that quarter. The data reveals a clear seasonal pattern, with peaks typically occurring in the second and third quarters of each year. However, following 2020, these seasonal highs begin to decline, and the overall trend shows progressively fewer listings. This suggests a post-pandemic market slowdown, potentially driven by reduced housing inventory, economic

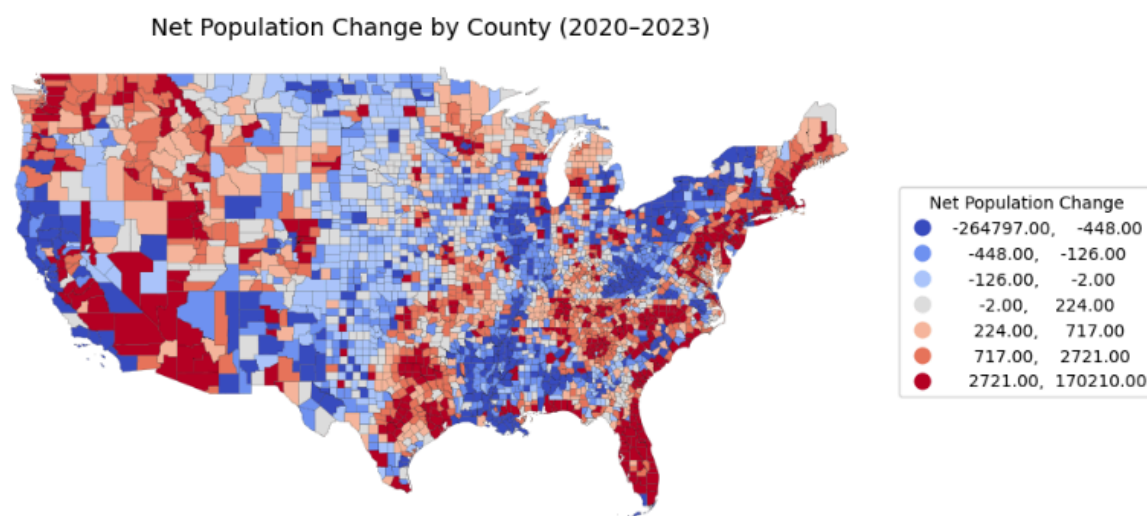
uncertainty, or shifts in seller behavior.



Exploratory Data Analysis for Research Question #2

Our second research question explores the impact of the COVID-19 pandemic: *“To what extent did the COVID-19 pandemic and its associated socio-economic shifts (including changes in migration patterns) lead to a significant and quantifiable change in average rental prices and/or rental price growth rates in major U.S. metropolitan areas, when comparing the pre-pandemic period (2018-2019) to the post-shutdown period (2020-2022)?”* To investigate this, we analyzed rental pricing trends across these two timeframes to identify where the most substantial changes occurred and what factors may have driven them. Our primary data source was the rentals subset, which included over 1.2 million rows of structured rental data, serving as the foundation for both our volatility and rent growth analyses.

To better understand the forces driving changes in rental prices, we incorporated U.S. Census population estimates to visualize net migration by county between 2020 and 2023. This map compares 2020 and 2023 population data to highlight where people moved during the post-COVID period. Areas in red represent higher net population growth, while areas in blue indicate population decline. Identifying these migration patterns helps explain where rental demand surged and where it may have contracted, offering geographic context to shifts in rent price and volatility across regions. There are some interesting phenomenon that we witness here. Some major metros lost large chunks of their population to the suburbs (Los Angeles, Minneapolis, Chicago, Seattle), while other large metros (Austin, Miami, Boston) saw large migration to their counties and surrounding counties.

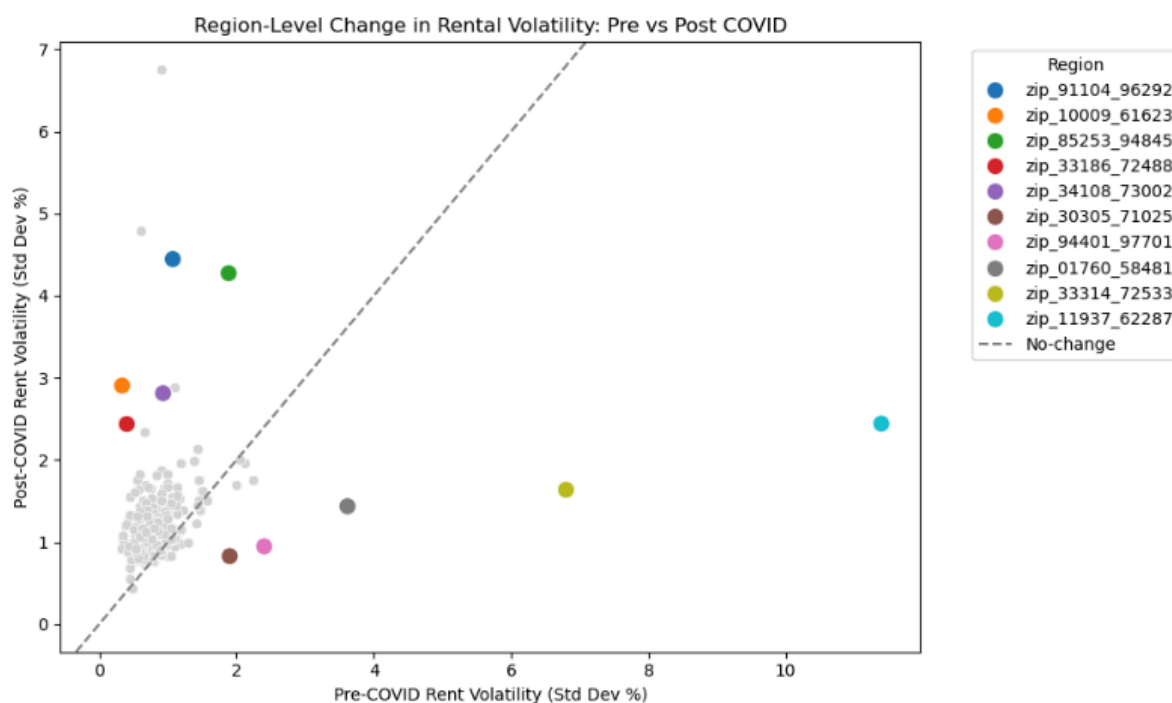


Net Population Change by county from 2020-2023

To explore Research Question 2, we examined how different factors influenced rental price growth in major U.S. metropolitan areas before and after the COVID-19 pandemic. We began by analyzing rental price volatility. Using average monthly rent values grouped by a unique region identifier, we labeled each data point as either “pre or “post” pandemic (based on

a March 01, 2020 cutoff). We then computed the standard deviation for each period to capture volatility shifts.

The scatterplot below compares the pre and post pandemic rent volatility across regions. Each dot represents a metropolitan area, with the diagonal line indicating no change. Regions above the line experienced increased volatility post-COVID, while those below were more stable. We had to identify the regions by a unique region identifier as there wasn't an accurate "States" column.

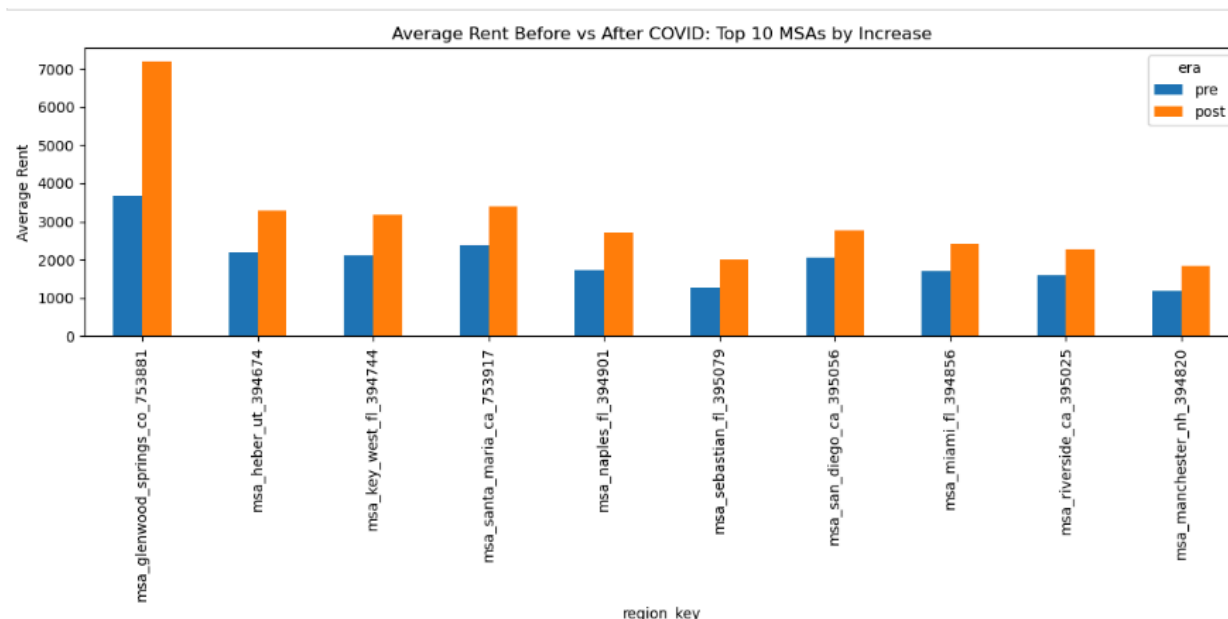


Pre and Post-Covid Volatility change by standard deviation percent.

(91104 - Pasadena,CA, 10009 - East Village, NYC,NY, 85253 - Maricopa County, AZ, 33186 - Miami, FL, 34108 - Naples,FL)
 (11937 - East Hampton, NY, Fort Lauderdale, FL, 01760 - Natick, MA, 94401 - San Mateo, CA, 30305 - Atlanta, GA)

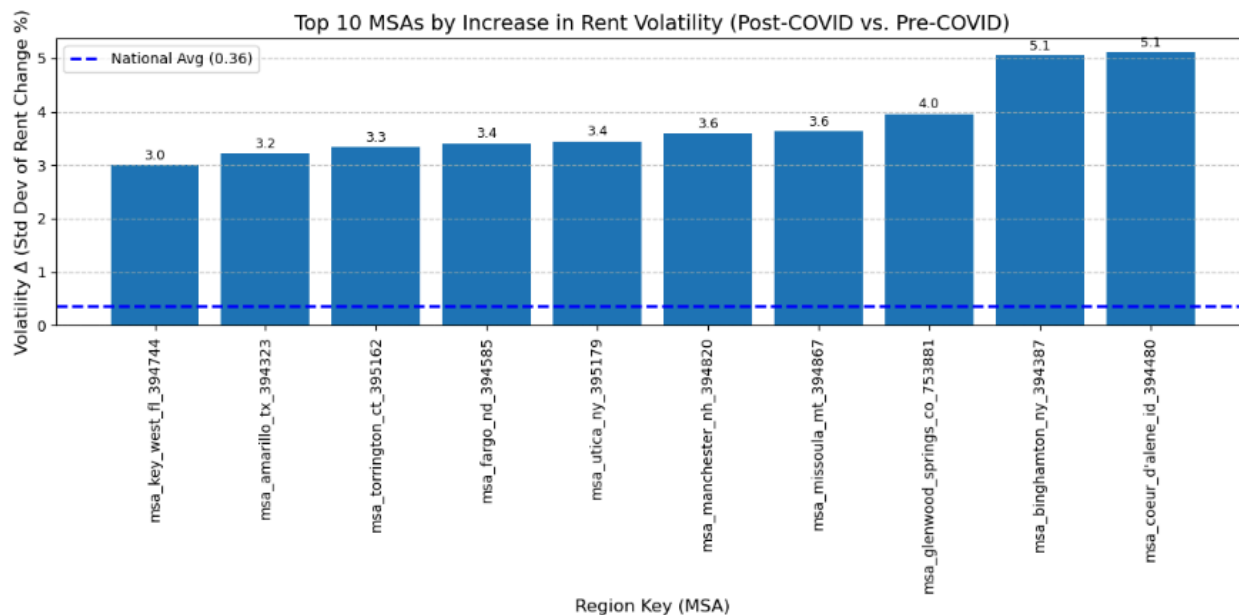
Following our volatility analysis, we sought to identify the metropolitan areas that experienced the largest absolute increases in rental prices after the onset of the COVID-19 pandemic. To do this, we calculated an average rent "delta" by subtracting the pre-pandemic

average rent (prior to March 1, 2020) from the post-pandemic average. This metric captures the raw change in dollar amount, allowing us to highlight the regions with the highest nominal growth in rent. The figure below displays the top 10 MSAs ranked by this measure.



Change in rent before vs. After Covid by top 10 MSA's. (By dollar amount)

Upon further analysis of rental price changes before and after the COVID-19 pandemic, we shifted our focus from absolute dollar increases to percentage-based changes in volatility. We determined that percentage growth offered a more meaningful comparison across markets of varying baseline rent levels. As shown in the figure below, the national average change in rent volatility was just 0.36%. In contrast, the top 10 metropolitan statistical areas (MSAs) experienced volatility increases ranging from 3.0% to 5.1%, highlighting significant regional disparities in post-pandemic rental market dynamics.



Top 10 MSAs by Increase in Rent Volatility (Pre vs. Post Covid)

Our analysis of rental price growth before and after the pandemic revealed several distinct patterns worth exploring further. First, while rent price volatility increased in some regions, others remained relatively stable. This will prompt us to investigate the characteristics shared by high-volatility areas. We also observed notable population shifts, particularly away from major coastal urban centers like New York and cities on the West Coast, with corresponding population gains in mid-sized cities across the interior U.S. Additionally, we identified substantial growth in suburban areas surrounding major metropolitan regions. These trends highlight a range of socio-economic dynamics that warrant deeper exploration in the next phase of our analysis.

While this report presents our initial exploratory analysis, we recognize the need for additional context to fully address Research Question 2. In the next phase of our project, we plan to incorporate supplementary datasets from the U.S. Census Bureau, particularly those related to migration patterns, population shifts, and socioeconomic indicators. These external

sources will provide deeper insight into the regional dynamics that may have contributed to changes in rent prices during and after the COVID-19 pandemic.

Key Findings and Limitations

In doing our exploratory data analysis we had a couple of different key findings that we aim to analyze further.

- **Home values vary drastically by region and density.** Densely populated areas, particularly in major metropolitan markets, consistently show higher home values compared to rural regions.
- **Housing prices accelerated post-COVID, especially in high-demand, low-supply markets.** Markets with low supply such as Breckenridge saw major increases in listing prices
- **Home type plays a role in value growth.** Condos and Co-Ops saw the highest percentage increases, suggesting that affordability pressure are pushing demand toward lower-cost housing.
- **Rental price volatility increased dramatically post-COVID.** Major MSA's experienced large swings in rent, highlighting supply constraints and pandemic-driven migration.
- **Supply-side constraints, are strong indicators of local market pressure.** Various markets saw increases in housing and rent prices as demand outpaced new supply in certain markets.

For limitations of our dataset we had some issues that affected the insight we could fully gain from our data.

- **Rentals dataset “State” column was incorrectly labeled.** The dataset had county listed under states and there was no one column designated for states. This made it harder to see which states had price growth
- **Some U.S. states are non-disclosure states.** This means they do not publicly report housing sale prices. This limits available and inconsistent data across states.
- **Not every state reports complete metrics.** Pending listings or days on market are skewed across different states.
- **Inconsistencies across subsets.** The zillow dataset although large, does have some discrepancies across the data subsets. This requires extra cleaning and caution when interpreting results

In conclusion, our analysis has given us some key insights into what type of data we have and is available. It shows what type of analysis and methodology we can use and how we can alter our data to get that information. We will include supplementary datasets to make our analysis more robust including, but not limited to, U.S. Census data on Poverty, Population Growth and Geo Spatial data. This will help to give us a better understanding of our data points and what the data is telling us.

Appendix:

1. Value Variables for Data Subsets (as listed on Data Card)

sales.parquet variables (with dtype):

- **Median sale price** (*float32*): Median closing sale price of homes sold in given region.
- **Mean sale to list Ratio** : Average ratio of sale price to original listing price. Values greater than one indicate selling over list price.
- **% sold above List/% sold below List**: Percentage of homes sold over and below listing price, respectively.

For_sale_listings.parquet Value variables: Description (dtype)

- **Median listing price** (*float32*): The median price at which homes across various geographies were listed.
- **Medial Listing Price (Smoothed)** (*float32*): The median price at which homes across various geographies were listed. (smoothed)
- **New listings** (*int32*): how many new listings have come on the market in a given month
- **New Pending** (*int32*): The count of listings that changed from for-sale to pending status on Zillow.com in a given time period.
- **Median Listing Price (smoothed)** (*float32*): Smoothed version of the listing price to account for volatility.

- **Home_values:** Mid Tier ZHVI (Smoothed) (Seasonally Adjusted): Zillow Home Value Index(ZHVI) for mid-tier homes, smoothed over time and adjusted for seasonal change.

Home_value_forecasts.parquet (dtype):

- **Month Over Month % (Smoothed) (float32):**
- **Quarter Over Quarter % (Smoothed) (float32):**
- **Year Over Year % (Smoothed) (float32):**
- **Month Over Month % (Raw) (float32):**
- **Quarter Over Quarter % (Raw) (float32):**
- **Year Over Year % (Raw) (float32):**

Rentals (Values) Variables: Description (dtype)

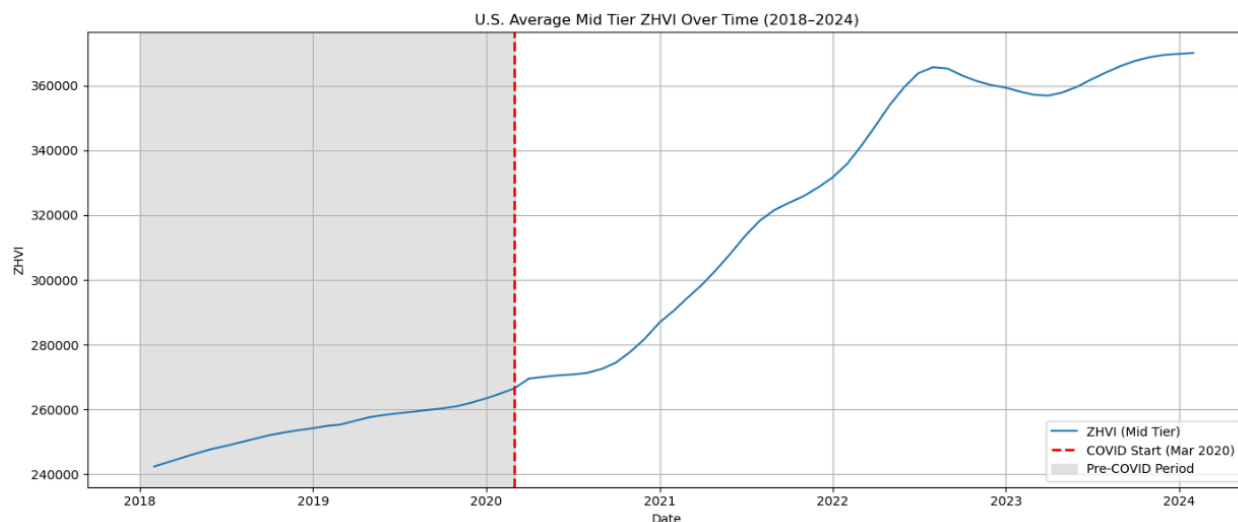
- **Rent (Smoothed) (float32):** Zillow Observed Rent Index (ZORI): A smoothed measure of the typical observed market rate rent across a given region. ZORI is a repeat-rent index that is weighted to the rental housing stock to ensure representativeness across the entire market, not just those homes currently listed for-rent. The index is dollar-denominated by computing the mean of listed rents that fall into the 40th to 60th percentile range for all homes and apartments in a given region, which is weighted to reflect the rental housing stock.
- **Rent (Smoothed) (Seasonally Adjusted) (float32):** Zillow Observed Rent Index (ZORI) :A smoothed measure of the typical observed market rate rent across a given region. ZORI is a repeat-rent index that is weighted to the rental housing stock to ensure representativeness across the entire market, not just those homes currently listed

for-rent. The index is dollar-denominated by computing the mean of listed rents that fall into the 40th to 60th percentile range for all homes and apartments in a given region, which is weighted to reflect the rental housing stock.

Days on Market Variables: Description (dtype):

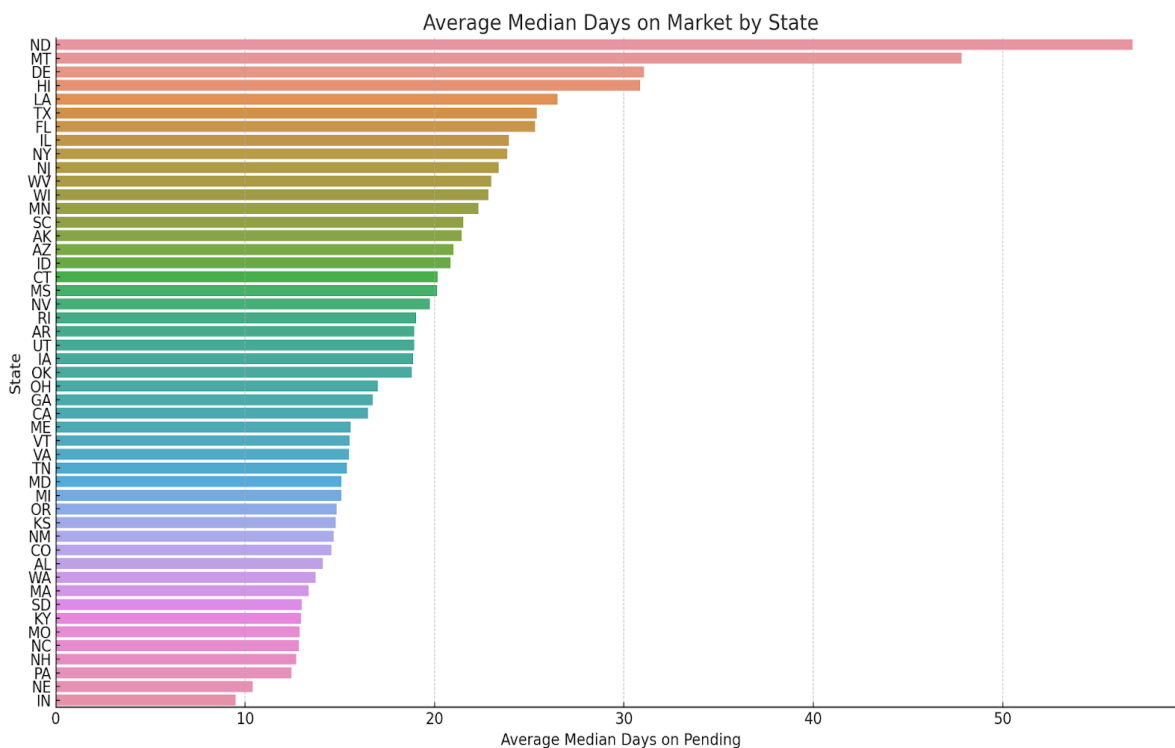
- **Mean Listings Price Cut Amount (Smoothed)** (*float32*)
- **Percent Listings Price Cut** (*float32*): The number of unique properties with a list price at the end of the month that's less than the list price at the beginning of the month, divided by the number of unique properties with an active listing at some point during the month.
- **Mean Listings Price Cut Amount** (*float32*):
- **Percent Listings Price Cut (Smoothed)** (*float32*):
- **Median Days on Pending (Smoothed)** (*float32*): median number of days it takes for homes in a region to change to pending status on Zillow.com after first being shown as for sale. (smoothed)
- **Median Days on Pending** (*float32*): median number of days it takes for homes in a region to change to pending status on Zillow.com after first being shown as for sale.

A.

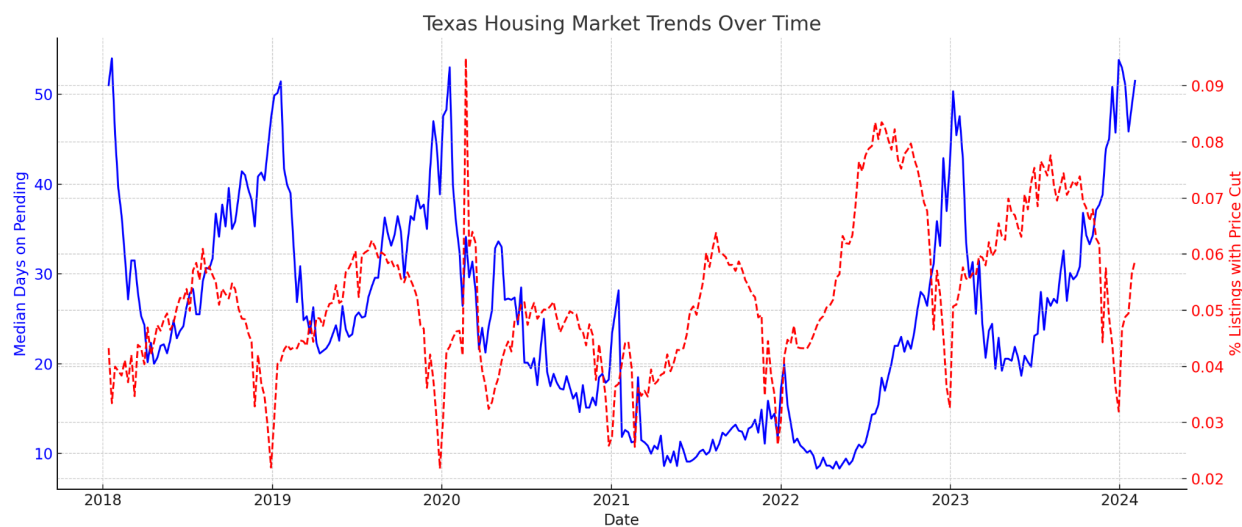


This is the zoomed in version of the previous graph but focuses on the years throughout 2018 to 2024, highlighting the gradual incline of ZHVI levels before 2020, followed by a steeper rise after mid 2020. The red dotted line indicates the time period when COVID-19 was declared as a pandemic by WHO. This graph highlights the continuous sharp curve throughout the years until in late 2022 when the incline finally levels. The visualization demonstrates the narrative of the acceleration of housing market prices during the COVID-19 pandemic, potentially driven by the fluctuations of a mixture of demand, migration, and limited supply of housing.

B.



C



Methodology

RQ1 - How do poverty rates and population density influence Zillow home value estimates across different regions?

- **Region level price modeling**

The analysis of housing value patterns are centered on estimating the impact of regional characteristics like poverty rate, population density, and local housing activity. These regions could be thought of as a record, described by features that depict socioeconomic stress and local market behavior.

Method 1: Multiple Linear Regression

A multiple linear regression model calculates the coefficients for each independent variable to gauge the change in effect on median home sale price in a region, while other variables are constant. This is a necessary model to serve as a baseline assessment of relationships between housing values and socioeconomic factors. For our context, the independent variables would be both poverty rate and population density, with the dependent(target) variable being the Zillow home value estimates(ex: ZHVI).

Specification: $ZHVI = \beta_0 + \beta_1 \times \text{Poverty Rate} + \beta_2 \times \text{Population Density} + \epsilon$

Using this model can enable us to effectively quantify the change of home values as poverty rate or population density changes. Additionally linear regression can be used to interpret the coefficients in a transparent manner, guiding our comprehension of whether different variables have a positive or negative impact on the target variable. A positive coefficient for population density would indicate that higher density areas would suggest higher home values and the negative coefficients would mean the opposite. This is true for all poverty as well.

Method 2: Ridge Regression

In order to mitigate multicollinearity issues of correlated variables (high/low population density leading to high/low listing activity), ridge regression takes multiple linear regression a step further by adding penalties for large coefficients. This will lead to increased stabilization of the model and its ability to fit to different regions.

Specification: Minimize $\sum (y_i - \hat{y}_i)^2 + \lambda \sum \beta_j^2$ where λ is the parameter that controls the intensity of the penalty. Using this model would allow us to stabilize the coefficients when adding more features and expanding the model.

Supplementary Method: Random Forest

In order to capture more complex, non linear relationships between variables, a random forest model will be used. This will help identify which regional features (poverty rate, population density) are most influential as pertains to housing prices.

RQ2 - To what extent did the COVID-19 pandemic and its associated socio-economic shifts (including changes in migration patterns) lead to a significant and quantifiable change in average rental prices and/or rental price growth rates in major U.S. metropolitan areas, when comparing the pre-pandemic period (2018–2019) to the post-shutdown period (2020–2022)?

Method 1: Multivariate Regression

Multivariate Regression is a statistical technique that is used to evaluate how combinations of different independent variables relate to a continuous dependent variable. It is different from multiple regression analysis used for question 1, as multivariate regression is not

limited to one output variable, whereas multiple regression only has a single dependent variable.

Using the Zillow Rent Indices(ZORI) data, multivariate regression can be used to identify how multiple variables such as population density, unemployment rate, housing supply trends etc., relate to a continuous Y variable, which would be the volatility change(post-pandemic minus pre-pandemic), or rent delta Since our research question contains more than a single or binary variable, and tends to focus on the combination of various COVID-19 migration factors(supply, volatility, price), it is crucial to utilize methods that are capable of efficiently identifying multiple variables. As our main goal is to explain the outcomes using complex features that include socioeconomic factors, we have determined Multivariate Regression would be ideal due to its flexibility with observational data.

ex) $\text{Rent Delta}_i = \beta_0 + \beta_1(\text{Population Density})_i + \beta_2(\text{supply trends})_i + \beta_3(\text{Unemployment})_i + \epsilon_i$

Method 2: Principal Component Analysis(PCA) + K-Means Clustering

Principal Component Analysis can be used to reduce the dimensionality of the data, breaking down correlated variables into uncorrelated sets called principal components. This method allows us to identify the directionality of the project and maximize variance in the data. Each of the principal components are linearly related to the original variables. For our purposes of this project, we can apply PCA to specific sets of features that might explain migration related trends including:

- WFH (Work From Home) percentage
- Population Density
- Pre, post-pandemic home affordability
- Unemployment rate changes
- Housing supply fluctuation
- Rent delta change

- Volatility change

PCA will reduce the dimensionality of these data to 2 or 3 principal components that would explain the most variance. The method will be followed by clustering approaches using K-Means or other appropriate clustering methods. Doing this enables us to identify region types that share similar trends in housing during the pandemic. The combination of PCA + clustering assists in revealing underlying patterns that may not be identifiable with regression analysis alone. By exploring the structural similarities and differences among distinct regions, it can efficiently aid our comprehension of how multi dimensional factors contribute to shaping the migration patterns and housing during the COVID-19 pandemic. If necessary, a combination of all of these methods mentioned above can be utilized to achieve an appropriate outcome.

In summary, the methodology that we intend to use contains a combination of various statistical and machine learning techniques to explore the complex dynamics of housing during and after the COVID-19 pandemic. For our first research question, the combination of multiple linear regression, ridge regression, and random forest enables us to explore the linear and non-linear relationships between poverty and population density with home value estimates. The second research question will introduce multivariate regression approaches with the support of PCA and clustering methods such as K-Means. Using these methods together, will assist in identifying non visible patterns that affect the migration related trends, ultimately guiding our project in identifying how socioeconomic factors have impacted the housing/rental market during and after the pandemic.

Data Visualizations and Analysis

Model Descriptions

Given the available data, we selected two subsets that best aligned with the focus of our analysis. Although both of our research questions centered around real estate, they explored distinct aspects of the market and required different modeling approaches. As a result, each dataset underwent its own tailored preprocessing and feature engineering to ensure the models we built were both meaningful and analytically sound.

For research question 1 *How do poverty rates and population density influence Zillow home value estimates across different regions?* We utilized population density and poverty rates of which were gathered from the US Census Bureau. These features were added to the working dataset to enhance its robustness and improve model insight. By integrating these socioeconomic indicators, we gained greater visibility into the relative importance of each factor in predicting home values.

For research question 2 *To what extent did the COVID-19 pandemic and its associated socio-economic shifts (including changes in migration patterns) lead to a significant and quantifiable change in average rental prices and/or rental price growth rates in major U.S. metropolitan areas, when comparing the pre-pandemic period (2018–2019) to the post-shutdown period (2020–2022)?* We expanded our dataset to include additional variables relevant to the pandemic's impact. These included unemployment rates, population figures, housing availability, and a COVID-19 stringency index. We engineered new features to enhance model performance and capture trends more effectively. For instance, we created an unemployment trend variable by calculating the change in unemployment from 2020 to 2022, providing a directional indicator of economic recovery. We also introduced a 3-month rolling window calculation for rent volatility to identify regions experiencing sharp upward or downward

rental trends. These feature engineering steps allowed us to enrich the dataset with previously unmodeled but meaningful dynamics.

Once the dataset was cleaned and formatted for analysis, we split it into training and testing subsets to build and evaluate our models. To address our first research question, we began with a Multiple Linear Regression (MLR) model as a baseline. After establishing this foundation, we implemented a Ridge Regression model to better understand how regularization influences model performance and parameter behavior. Finally, we benchmarked these models against more advanced algorithms, including Gradient Boosting, XGBoost, LightGBM, and Random Forest, to compare predictive accuracy and robustness.

For our second research question, we began with a multivariate regression model as a baseline. We refined the model through several iterations, removing features that added noise or lacked predictive value. After completing the regression analysis, we shifted to unsupervised learning by performing Principal Component Analysis (PCA) followed by K-Means clustering. To determine the optimal number of clusters, we used both the silhouette score and elbow method, which indicated that three clusters provided the best balance before the gains in clustering performance began to level off.

Model Results and Analysis

Research Question 1:

How do poverty rates and population density influence Zillow home value estimates across different regions?

For our first research question we decided on doing a log transformed-Multiple Linear Regression analysis as the base model. In order to compare the results we utilized the following models: Ridge Regression, Gradient Boosting, XG Boost, LightGBM and Random Forest (of which we tuned). This gave us a more diverse array of metrics to grade how the model performed. After running this we had multiple R^2 and Root Mean Squared Errors (RMSE) that gave us a clear view of how well we did. The Multiple Linear Regression base mode had a RMSE of .626 and a R^2 of 1.10% at the city level and at the county level it had a RMSE of .508 and a R^2 of 33.30%. We saw across the models that the county level data performed markedly better in both metrics. We compiled both of these figures for county and city level data across the models in the tables below.

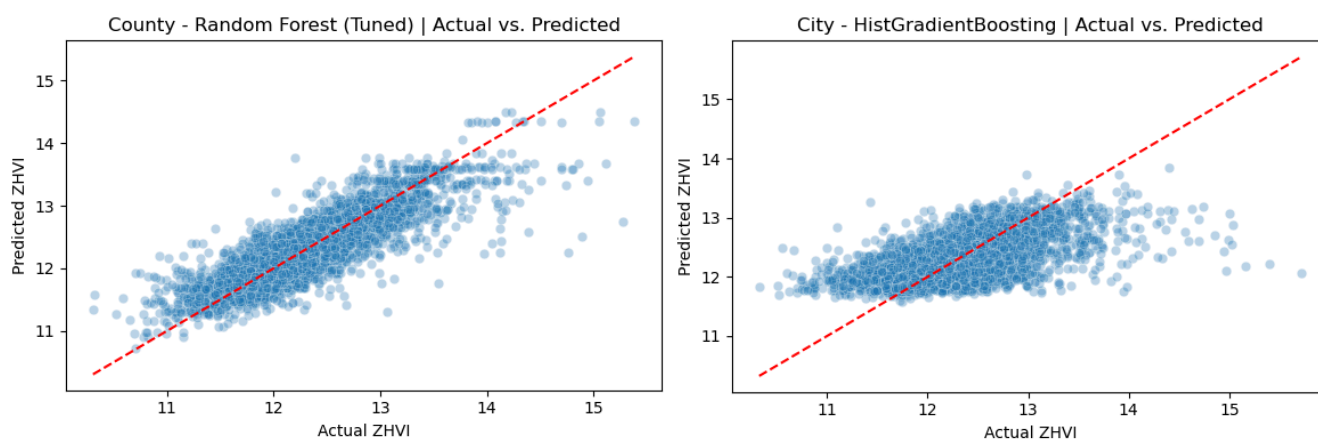
For our city level metrics, the best performing model when it came to overall R^2 metric was the HistGradientBoosting model that we ran. This model had a R^2 of 34.96% meaning it explains 34.96% of the variance in the housing value data.

For the county level metrics, the best performing model was the Random Forest that was hyper parameter tuned. The RMSE was at .3298 and the R^2 was 71.91% showing that the model was able to explain 71.91% of the variance in housing values. This is a significant leap from the Linear Regression and Ridge Regression models that were originally used.

City Level Metrics		
Model	RMSE	R ²
Linear Regression	.6257	1.10%
Ridge Regression	.6257	1.10%
HistGradientBoosting	.5074	34.96%
XGBoost	.5093	34.47%
LightGBM	.5086	34.64%
Random Forest (Tuned)	.5250	30.37%

County Level Metrics		
Model	RMSE	R ²
Linear Regression	.5082	33.31%
Ridge Regression	.5082	33.31%
HistGradientBoosting	.3761	63.47%
XGBoost	.3699	64.66%
LightGBM	.3741	63.85%
Random Forest (Tuned)	.3298	71.91%

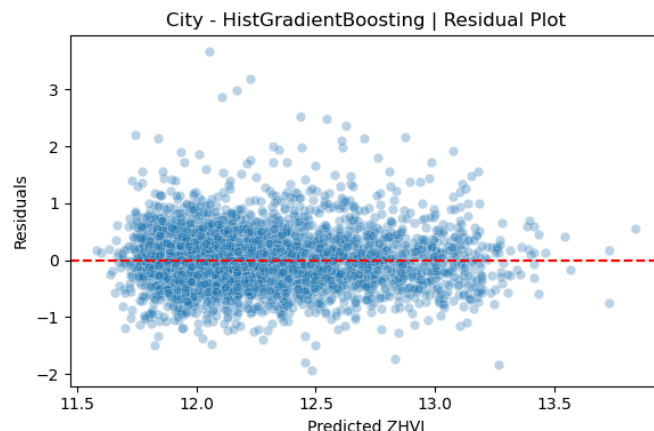
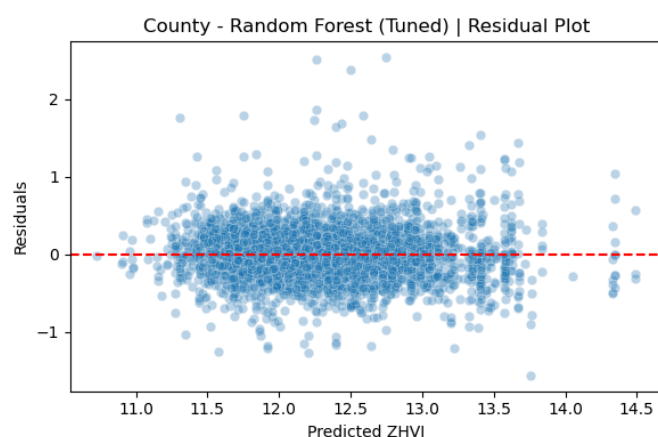
To gain further insight into how these models performed we looked at various data visualizations to see how the data lined up in select models. We first wanted to take a look at the two best performing models on a scatterplot. This would help to enhance how we understand the data and what the metrics look like visually plotted.



The two scatter plots above compare actual versus predicted ZHVI values for county-level(Random Forest) and city-level(HistGradientBoosting) models. In the county-level plot, predictions are aligned closely with the actual values along the red diagonal line, demonstrating that the model captures regional housing trends effectively. This strong performance suggests that the Random Forest model is suited well for modeling the aggregated county data, where broader patterns are more stable.

In contrast, the city-level model shows a larger deviation from the diagonal line, with predictions showing a tendency of underestimating higher ZHVI values. This outcome indicates weaker performance and highlights the challenge of modeling localized urban markets, where price driving variables are more complex. While the model is still able to capture the general trend, it is clear that it lacks precision at extremes, such as high-cost areas.

The residual plots further emphasize the difference between the two models across region levels. At the county-level, the residuals are relatively evenly distributed, centered around zero. This indicates that the model's errors are

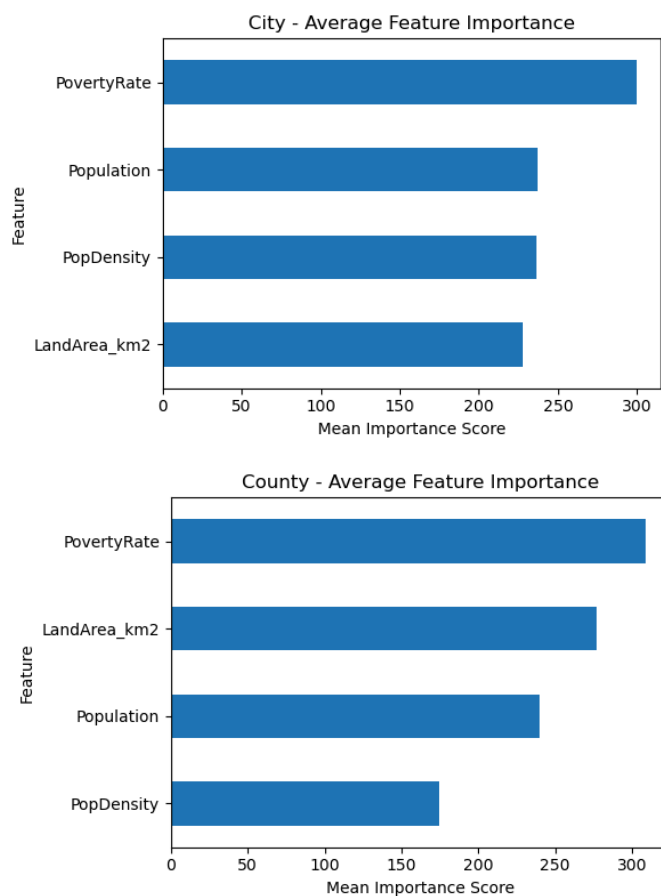


well balanced and consistent across most counties. Although there are a few outliers especially in higher-value regions, they are not strong enough to interfere with the model's stability.

On the city-level, the residuals are more broadly distributed, with error margins larger than the random forest, especially in higher predicted ZHVI values. This suggests that the city model had more difficulty in making accurate predictions in higher value urban areas, likely due to an increased market variability. Overall we can highlight that the city-level model has limited generalizability in finer settings.

Next, we aimed to identify which metrics had the greatest impact on our models. By understanding how specific variables influence housing values, we could provide more targeted and actionable recommendations to individual regions for improving housing affordability. To accomplish this, we introduced the Mean Importance Score; a consolidated metric that

quantifies the relative importance of each feature in the model, helping us determine which variables carry the most predictive weight.



These side by side bar charts show the average importance scores of four key features aggregated across models for both county and city:

- Poverty Rate
- Population
- Population Density
- Land Area

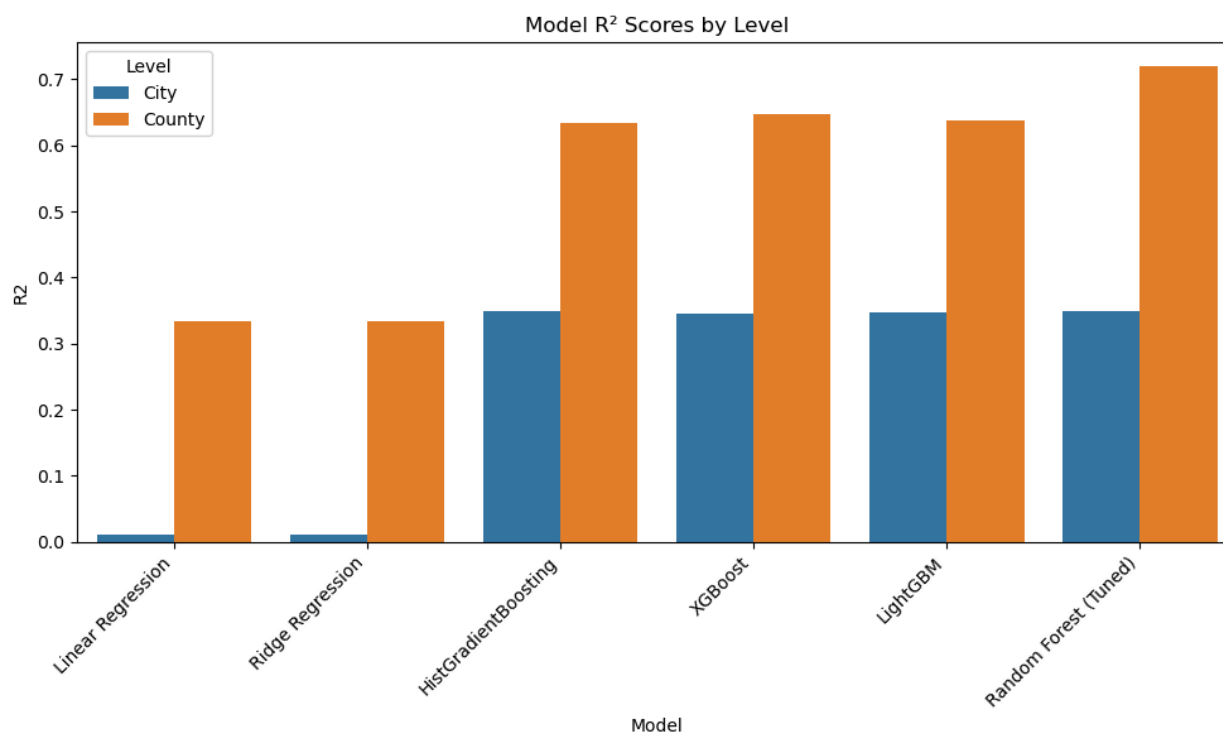
Key Observations

In the **county-level model**, poverty rate stands out as the most impactful predictor of home values. This finding reinforces earlier analyses: counties with higher poverty levels generally have lower median home values. Land area ranks as the second most influential feature in the county-model. The reason for this could be because larger counties often contain both rural and suburban areas, introducing a wider range of characteristics that impact price trends. Population is also relatively higher ranked, suggesting that the demand of housing is reflected in how many people reside in the region. Meanwhile, our hypothetical main predictor, population density appears to have the least amount of impact, at least in the county level. This may be due to the density being not as influential and informative when dispersed in larger and diverse regions.

At the **city level**, the mean feature importance scores are slightly different than the county-level model. Poverty rate still remains as the dominant predictor, showing consistent relevance regardless of the geographic scale. However, unlike in the county model, population and population density show nearly equal feature importance. This suggests that in urban housing markets, both factors of how many people live and how closely they live together is significant in influencing price behavior. In the city level, land area is relatively less significant, likely because cities generally vary less in geographic size, but more in economic, structural characteristics.

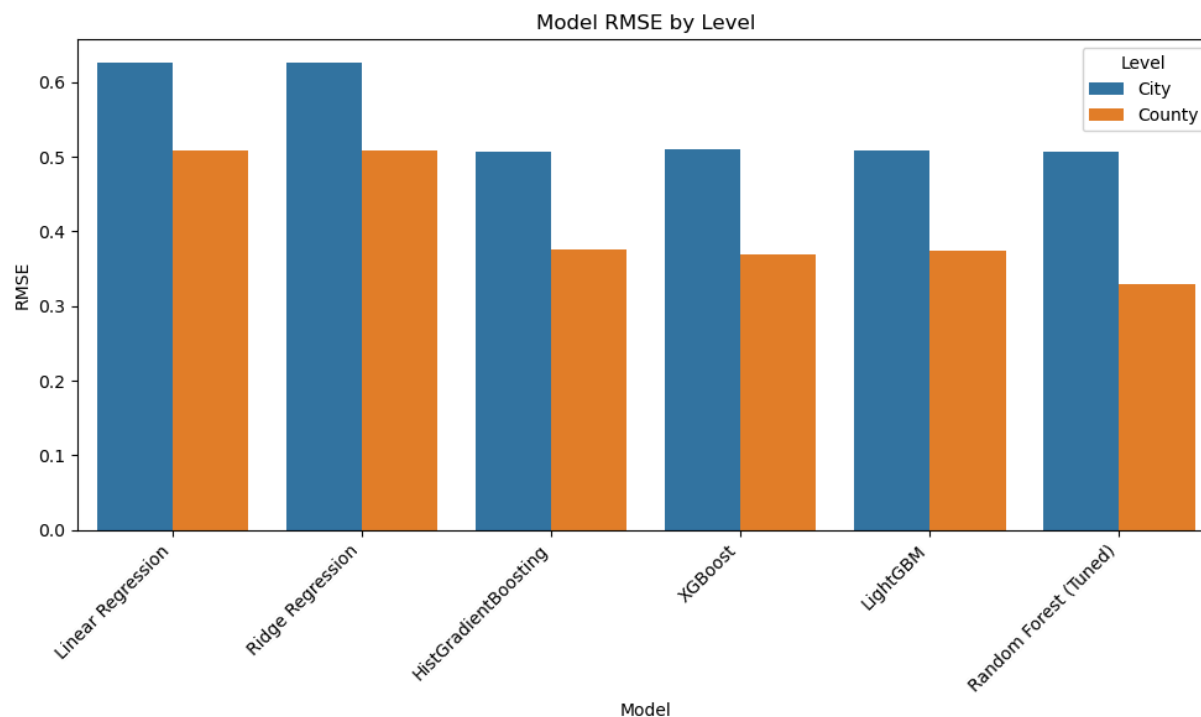
In summary, while poverty rate is a reliable predictor at both levels, other features vary in importance depending on the scale. More larger county-level predictors rely more heavily on broader variables such as land area and population, whereas city-level models prioritize density and concentration of population.

Lastly, we visualized the performance of each model at both the city and county levels. This comparison allowed us to assess how closely the models aligned across different geographic scales and evaluate how each model performed relative to one another.



In terms of R^2 scores, the results clearly show that the models perform significantly better on county-level data. While our initial methods of Linear and Ridge regression failed to explain a negligible portion of variance at either region level, tree-based models such as HistGradientBoosting, XGBoost, LightGBM, and tuned Random Forest achieve R^2 scores of around 0.65 to 0.7 at the county level.

In contrast, these identical models achieve only about 0.35 to 0.4 at the city level. This suggests that county level housing prices are better in predictability, likely due to aggregated data smoothing out the local variability and highlighting broader socioeconomic patterns.



When it comes to the RMSE, the comparison above further explains similar conclusions. We can observe that at the city level, every single model shows higher RMSE, implying that the average prediction error is greater than the county-level prediction. RMSE values are significantly lower at the county level, with boosting models and tuned Random Forest showing the least errors. Similarly to the R^2 scores mentioned above, both regression models (linear & ridge) show the poorest performance in RMSE, indicating their limited ability to predict non-linear relationships in the data.

To summarize, tree-based models, particularly Random Forest and gradient boosting algorithms consistently outperform linear models. Regardless of the method used, county level predictions show higher accuracy and stability than at the city level. These observations indicate that housing market behavior has higher predictability at broader geographic scales, where complex local dynamics are smoothed out. Therefore we can make the conclusion that broader socioeconomic indicators have stronger and stable effects to the overall housing predictability.

Model Results and Analysis

Research Question 2:

To what extent did the COVID-19 pandemic and its associated socio-economic shifts (including changes in migration patterns) lead to a significant and quantifiable change in average rental prices and/or rental price growth rates in major U.S. metropolitan areas, when comparing the pre-pandemic period (2018–2019) to the post-shutdown period (2020–2022)?

For research question 2 we ran a log transformed-Multivariate Regression model taking multiple features that were given and engineered as well. In doing so we were able to get a base model for our regression analysis. The log transformed Root Mean Squared Error (RMSE) is 0.28. When exponentiated ($e^{0.28}$), this suggests that, on average, our predictions tend to deviate from the actual rent values by a multiplicative factor of approximately 1.32. This means that, for a given rent, the predicted value is typically within a range of being about 32% higher or 24% lower than the actual rent. The R^2 of .3356 tells us that about 34% of the model can be explained by our metrics. The formula that was rendered from the regression analysis can be found below.

These results align with the general understanding that real estate models, while influenced by macro trends, are highly regional. Our model captures broader patterns affecting rental prices across the nation, but it acknowledges that localized micro trends and complexities are not fully captured by large-scale models, leading to inherent variability in predictions.

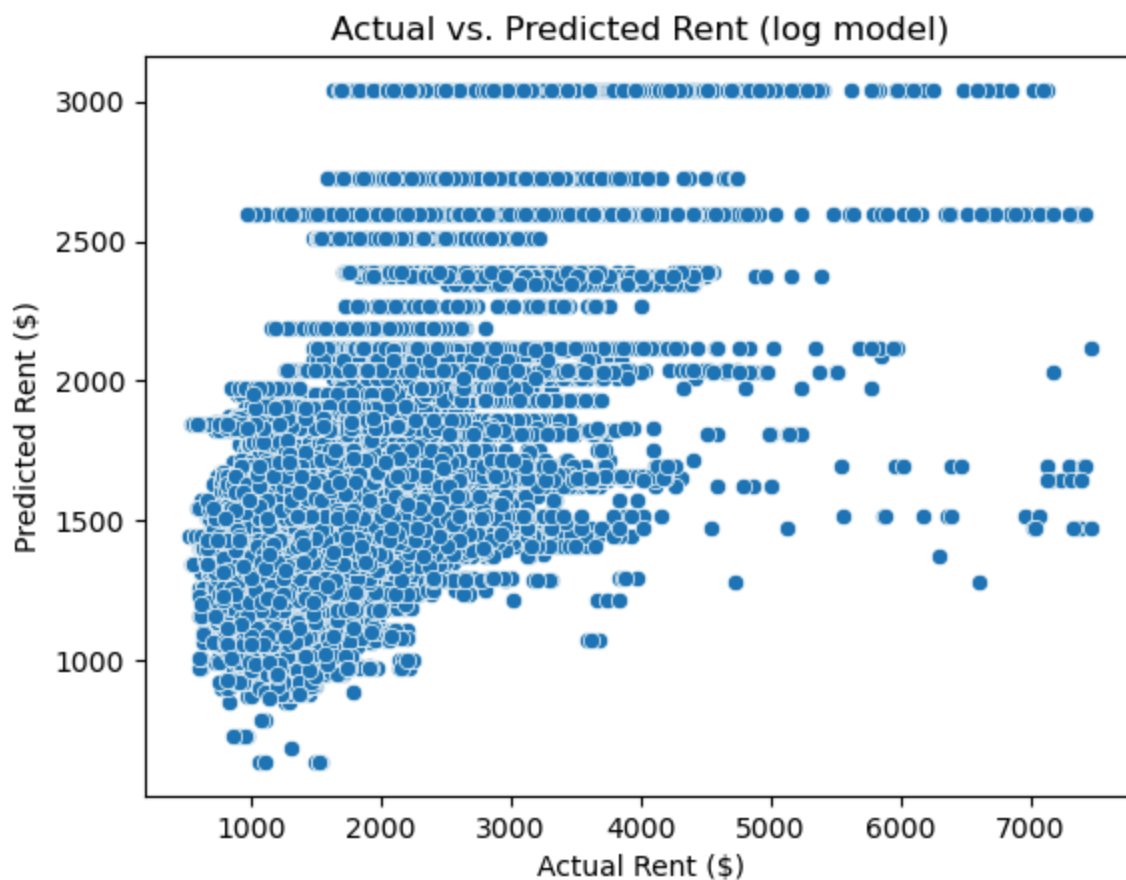
Root Mean Squared Error (exponentiated)	R^2
0.28 (1.32)	33.56%

Multivariate Regression Formula:

$$\log(\text{rent}) = 7.3886$$

$$\begin{aligned} &+ 0.1127 \times \text{Stringency_Mean} \\ &+ 0.2593 \times \text{pop_density} \\ &- 0.0235 \times \text{housing_unit_growth} \\ &+ 0.0412 \times \text{housing_unit_pct_growth} \\ &- 0.1780 \times \text{housing_unit_density} \\ &- 0.1159 \times \text{unemp_trend} \\ &- 0.1394 \times \text{unemp_mean} \\ &- 0.0868 \times \text{pop_change} \\ &+ 0.0101 \times \text{pop_pct_change} \\ &+ 0.0241 \times \text{pop_avg} \\ &+ 0.0782 \times \log_pop_avg \end{aligned}$$

After training the model, we visualized its performance by comparing predicted rent values to actual rent prices. As shown in the scatterplot below, the majority of data points fall in the lower-left quadrant, reflecting that most rental activity occurs in more affordable regions. To improve model accuracy and reduce skewness caused by extreme outliers, we limited the dataset to properties with actual rents below \$7,500 per month. This adjustment helped create a more balanced and representative model.



This scatter-plot compares predicted rent (vertical axis) against the actual rent in dollars (horizontal axis), based on a log-transformed linear model.

- Core market (< \$2 000 actual rent).

Most observations sit below \$2 000, so we see a thick cluster here. Predicted values range roughly from \$800 to \$1 800, following the overall trend but—not surprisingly for real-world data—scattering around rather than falling on a perfect 45-degree line.

- “Shelf” at about \$3 000 predicted rent.

Above the core range the points flatten into a horizontal band. Because we trained on the log of rent and then converted back to dollars, very large log values compress when

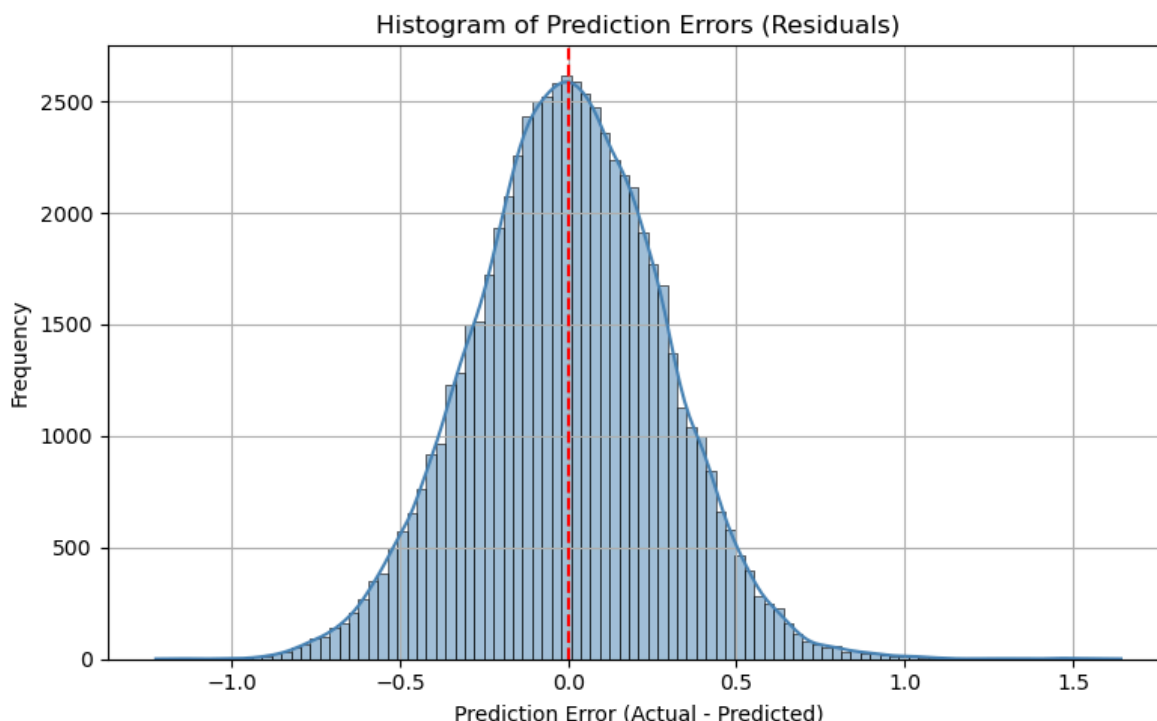
exponentiated, effectively capping many predictions near \$3 000.

- Systematic under-prediction of high-end units.

For true rents above roughly \$4 000 the cloud lies well below the ideal diagonal, showing the model predicts only \$2 000–\$3 000 where actual rents can exceed \$6 000 or \$7 000.

In short, the log-transformed model tracks everyday rents reasonably well but flattens out for luxury-level prices, consistently under-estimating the most expensive markets.

Next, we evaluated the model's accuracy by analyzing a histogram of the prediction errors. This approach provided a clear and detailed view of how well the model performed at scale, allowing us to assess whether it consistently overpredicted, underpredicted, or stayed balanced across the dataset.



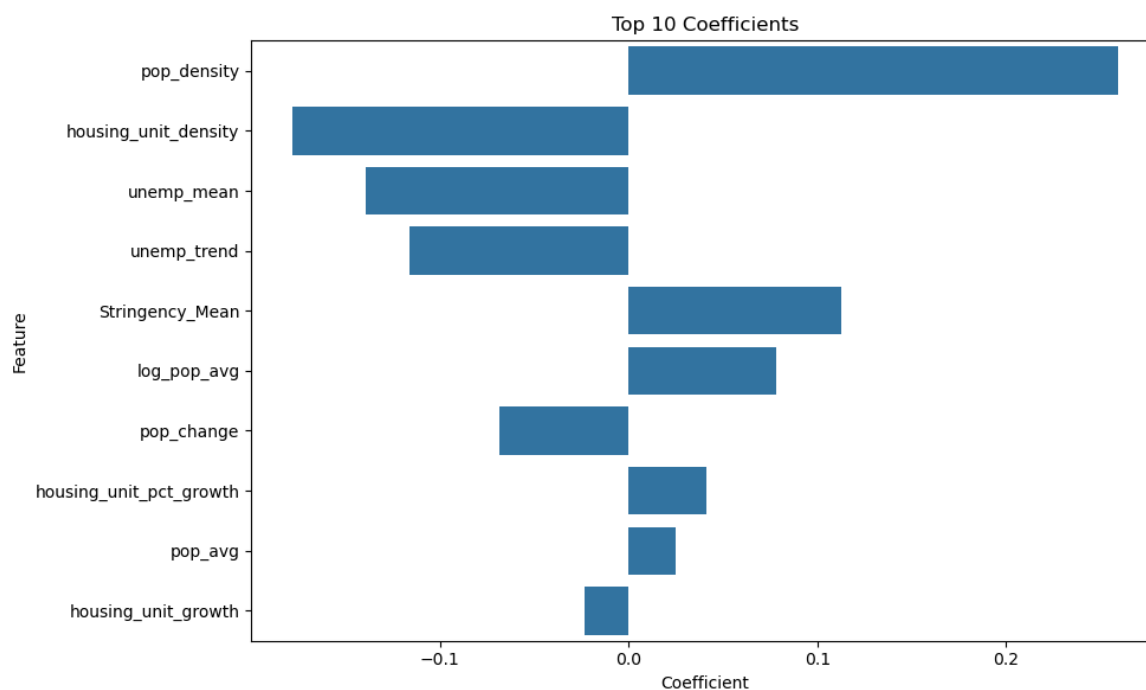
This histogram of prediction errors (residuals) shows a classic bell-shaped distribution, peaking sharply around zero. This indicates that the model tends to underpredict and overpredict with roughly equal frequency, suggesting a well-balanced and unbiased regression model.

The narrow concentration around zero means that for the vast majority of counties, the predicted rent is very close to the actual rent, a strong signal of overall model reliability. The symmetry of the curve supports the idea that residuals are distributed normally, which is a good sign in terms of regression assumptions like homoscedasticity.

However, we also see light tails extending to both the left and right, indicating the presence of some large prediction errors. These outliers likely correspond to counties with extreme rent values — either very high-end or unusually low-cost areas, where the model has less training data or explanatory power.

In short, this distribution suggests that the model is accurate and consistent for most cases, but there's still room for improvement in handling edge cases or outlier regions.

Lastly for our Regression model we wanted to see which factors weighed the most when it came to the regression model. In a regression model we can hypothesize what we believe would be key factors, but until we run the model we actually don't know. Our hypothesis was that we believed that stricter COVID guidelines would be negatively calibrated to rent growth. After tuning the regression model we found the following factors were key in determining rent growth metrics.



This bar chart highlights the top 10 most influential features in our linear regression model predicting rent prices. Each bar represents the coefficient value of a feature, showing both the magnitude and direction of its impact.

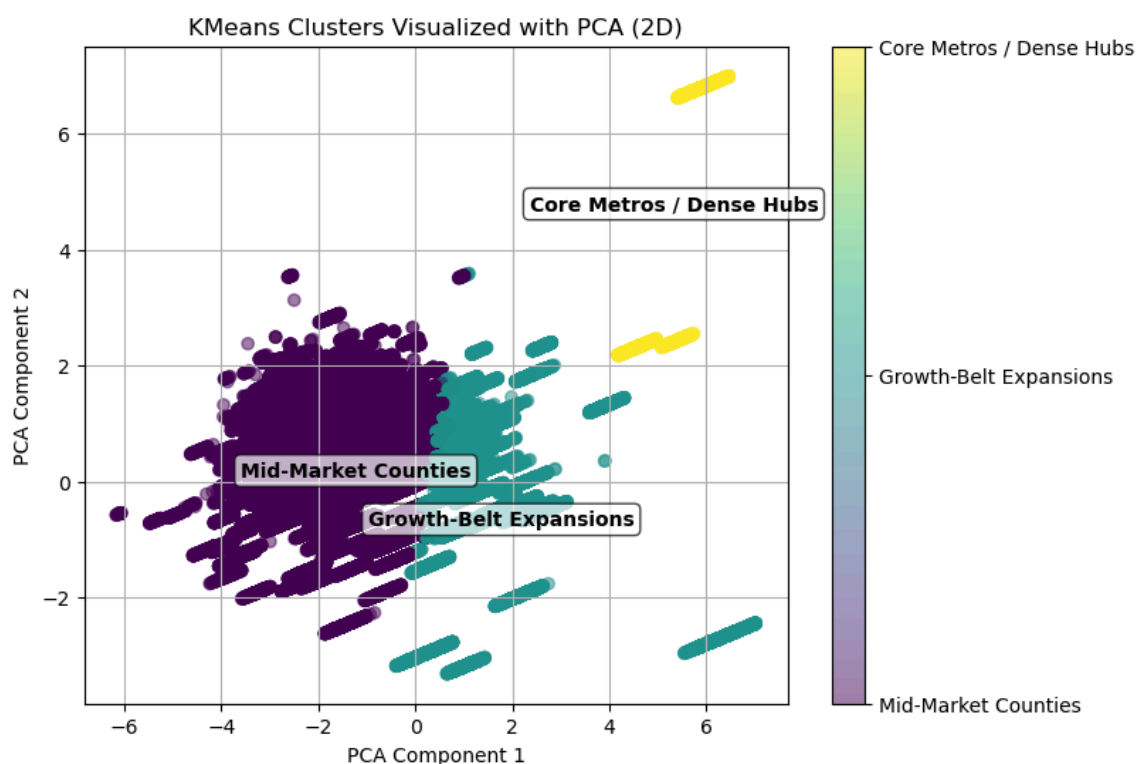
- On the positive side, features like `pop_density`, `Stringency_Mean` and `log_pop_avg` have strong positive coefficients, meaning that as these values increase, so does the predicted rent. This aligns with expectations, areas with higher population density or more volatile rent changes tend to have higher rental costs.
- On the negative side, `housing_unit_density`, `unemp_mean` and `unemp_trend` show strong negative associations. This suggests that areas with higher recent rent growth or higher average unemployment rates are associated with lower current rents, potentially indicating market corrections or affordability ceilings being hit.

A particularly noteworthy insight is the positive coefficient for `Stringency_Mean`, which captures the average COVID-19 policy strictness in a state. This runs counter to our original hypothesis — we expected more stringent regions to suppress rental growth due to restricted mobility or economic slowdowns. However, this result may reflect underlying confounders: many of the most stringent states (e.g., California, New York) also have structurally high rents and demand regardless of policy.

Overall, this plot provides a clear snapshot of the relative weight each factor contributes to rent prediction. It also surfaces valuable questions for further investigation, such as why certain socioeconomic trends (e.g., population shifts or unemployment patterns) are influencing rent outcomes in unexpected ways.

After completing our regression analysis, we proceeded with K-means clustering and Principal Component Analysis (PCA) to classify regions based on shared characteristics. While it might be intuitive to group cities like New York and Los Angeles together, these methods allowed us to uncover deeper, data-driven similarities among regions. PCA was first used to

identify the top contributing factors driving regional classification (see Appendix A). We then applied K-means clustering and used the silhouette score (Appendix B) to determine the optimal number of clusters, which was found to be three.



Principal Component Analysis (PCA) was applied for the reduction of dimensionality of our multivariate dataset. This allowed us to project high dimensional data (unemployment rate, housing supply, pandemic policy stringency) into a 2D space for the ease of visualization. PCA was followed by KMeans clustering, to categorize counties into three distinct groups. These clusters were able to reflect unique regional profiles based on structural similarities.

Growth_Belt Expansions include areas such as Las Vegas, Austin, Nashville, and Miami. These counties experienced rapid in-migration, rent increase, and housing supply expansion.

Mid-Market Counties represent the cities with moderate population density, moderate housing unit metrics, and lower stringency (this is a big part of the dataset).

Core Metros/Dense Hubs are the cities with high covid stringency, high unemployment, and are dense (Los Angeles, New York, Seattle).

These groupings demonstrate a framework for interpreting how rent trends differ not just by region, but by shared socioeconomic structures.

Overall Conclusion

Our analysis was guided by two core research questions: first, to determine how poverty and population density affect Zillow Home Value Index (ZHVI) across regions, and second, to assess how the COVID-19 pandemic and associated socio-economic shifts impacted rental prices in major U.S. metros. To address these, we employed a blend of supervised learning techniques, including multiple linear regression and tree-based models, as well as unsupervised learning via PCA and K-Means clustering to uncover structural patterns in housing behavior.

For Research Question 1, our findings showed that poverty rate was a consistently strong negative predictor of home values, especially at the county level. Population and land area also emerged as important features, with non-linear tree models like Random Forest and Gradient Boosting clearly outperforming linear regression approaches. Notably, we found that model performance was substantially stronger at the county level than at the city level, highlighting the importance of scale in housing data analysis.

For Research Question 2, we log-transformed rent to reduce skew and ran a multivariate regression, which explained 33.56% of variance (RMSE = 0.28). While we hypothesized that more relaxed COVID restrictions would correlate with greater rent growth, the opposite

emerged: more stringent states (e.g., California, New York) showed higher rents, likely due to existing structural demand. Features like population density and average population pushed rents upward, while housing unit density, unemployment trends, and population decline suppressed them. PCA and clustering further revealed three distinct regional patterns: Growth Belt expansions, mid-market counties, and dense urban hubs.

In summary, our project demonstrated the nuanced and multifactorial nature of housing dynamics in the U.S. Using a combination of regression, ensemble models, and clustering, we provided a layered perspective on how different features interact with housing prices and rent growth across regions. These insights can support future research and inform urban policy, especially in planning for resilient and equitable housing strategies post-COVID.

Ethical Recommendations

In the world of data ethics, predictive analysis doesn't just track markets, markets are also heavily influenced by them. The use of machine learning models in home price estimation or migration trends analysis have far reaching ethical and moral consequences: notably regarding bias, fairness, and social impact. Algorithms have clear advantages in that they can remove human bias from analysis. But algorithms that are created from socioeconomic factors and housing prediction models, if used without oversight in any manner, could reinforce institutionalized inequality or produce false forecasts that have real-world impacts on the lives and decisions of individuals.

The core of this study uses models that account for population density, poverty rates, rental history, and neighborhood demographics. These are strongly tied to previous trends of discrimination, redlining, and housing inequality in the United States. Predictive models may then end up capturing and even increasing these disparities by including systematically lower estimates in historically underprivileged neighborhoods, regardless of actual market trends. The data can inherit the prejudices of prior decisions in ways that might not reflect the impact that they have had on marginalized groups (Selbst, 2016). This becomes impactful on a macro level when data can potentially impact the actions of the driving decision markets of the industry, such as federal reserve interest rates or large real estate investors.

The data from this project highlights the changes from the COVID 19 virus and its widespread impacts on the real estate market. It is important to remember that COVID was an exceptional time, and although unique analysis can be gained for future pandemic or outlier events, they are not the standard protocol for real estate markets. Population migration models derived from COVID-era experiences might misinterpret short-run behavioral accommodations as trends. Without context-sensitive modeling, there is a risk that communities that were economically affected more severely or that had tighter public health restrictions might be unfairly penalized in

the system of valuation. Even if the valuation is accurate for the time, it could give investors a false sense of socio economic shifts on a macro scale. For example, during COVID, rents in high-density areas had fallen more than 10 percent while suburban/rural areas saw property demand and values rise, but on a larger timescale, the opposite trend may be happening (Ramani, 2021).

The underlying ethical considerations between these research questions is the use of computer appraisal models like the Zestimate impacting public perceptions of home value and are often used by owners, buyers, and sellers to make life-changing decisions. Any model based on projects like this must transparently report inputs and limitations. A model may be statistically valid, yet if its assumptions or data bias are not made transparent, it could deceive users, or distort their overview with false perceptions. This entails being open about what factors are correlational and what factors are predictive. For instance, showing poverty rate to be correlational with lower Zestimate value is not necessarily showing that poverty results in low value—it may reflect broader social structures.

Without openness, users could make incorrect and harmful inferences. Our initiative works with publicly available information from Zillow and Census.gov. Although these are anonymized and aggregated, there remains an obligation not to use data in a fashion that can be traced back to individuals or neighborhoods in a manner that can promote exclusionary practice, such as investors avoiding disadvantaged areas purely based on model predictions. Caution must be taken over contextualizing and anonymizing visualizations and outputs used in public or commercial purposes.

Real estate ethical data science requires stringent examination of social consequences. Though the output of this project is significant regarding home value drivers and migration, its results must be interpreted with humility and restraint. Zillow and similar websites owe it to themselves and others to make their tools work for housing equity and transparency, and for prediction accuracy. Analysts must constantly go back to reviewing how algorithmic decisions

impact individuals and communities—and make models more of a guide than gatekeeper to opportunity.

Challenges

One of the earliest challenges was implementing data from different sources. Initially, the data that we got from HuggingFace, included seven different subsets, those included Zillow's Home Value Index(ZHVI), ZORI, New construction, etc. Many of these datasets varied in both structure and granularity. It was challenging to figure out which of these datasets were going to be mainly relevant to our purposes of the project. Additionally, splitting the datasets for four people to work on was not the smoothest process, as each of us may have had different perspectives and information extracted from the datasets.

An additional challenge regarding the datasets was the amount of missing data and temporal gaps. For instance, some housing variables such as co-op units had significant gaps in certain years and states. This caused a dilemma, and implementing could potentially introduce bias or falsified trends, while removing them entirely, would be riskful in its own way, as we could lose meaningful information. Ultimately, careful trade-offs had to be made in order to maintain reliability.

One of the hardest things to overcome was the realization after finishing up the EDA for research question one, that our sales.parquet data did not include sales data outside of key metropolitan areas of the United States. The *Region* field represented broad metropolitan areas despite having city or town names. This led to a hard pivot away from that dataset, and the use of ZHVI data straight from Zillow which included city and county identifiers, but lacked key features, including home types.

Looking back, we could have restated our research question to align more with our new understanding of the available data and model around metropolitan areas instead of city and county. ([Appendix](#)) While we were able to make the switch to ZHVI data, this early misstep drastically limited our available features for modeling and added considerable man-hours.

A huge part of our research explored several different geographic scales of housing trends. While this allowed us to explore urban versus rural housing market trends, it made it difficult to make generalized conclusions about localized urban development or migration patterns. Although county-level and city-level data did enable us to make more detailed and accurate comparisons between different geographic regions, the datasets we used limited us to strongly conclude our findings.

Recommendations and Next Steps

Conclusions:

Throughout this project, we were able to gain valuable insights into how a combination of housing, and socioeconomic data can be used to understand regional housing trends across the United States. By thoroughly analyzing Zillow's datasets alongside socioeconomic data such as poverty, population density, land area etc., we observed clear distinctions in market behavior between urban and rural regions, as well as observable shifts influenced by the COVID-19 pandemic.

Recommendations for Future Research and next steps:

1. Utilize More Granular Geographic Data

Our analysis benefited significantly from the use of county and city level data, with the implementation of FIPS codes and sub-state geographic codes. To better capture the nuance between urban and rural housing trends, we recommend future studies continue to use and implement more specified ZIP code level or county level data. This would allow researchers to identify a better and more precise spatial interpretation. This approach can assist researchers dive deeper into studying local housing and development patterns, and analyze population shifts with far greater precision.

2. Expand Feature Set with Socioeconomic Variables

While our analysis did include socioeconomic variables such as poverty, population density, and population etc. it would be beneficial in future research to expand on these variables. Introducing a more broader range of these variables, such as:

- Median household income and poverty rate - Exploring and contextualizing affordability
- Unemployment rate and job growth - For tracking the economic growth and health
- Rental costs, homeownership rates - For understanding home price trends
- Demographic age profiles - For better understanding of who is migrating and the reason

3. Deepen Analysis of COVID-19's Impact

While we briefly addressed the influence of the pandemic, further research should isolate COVID-specific migration trends using more direct data sources. These sources can include data from USPS(change of address) or other relevant movement datasets. This would allow future research to identify population flow changes, creating a clearer picture of how the pandemic changed and solidified certain housing markets. Additionally, comparing the population growth between pre and post pandemic eras in different geographical scales can help identify long-term structural changes in regional housing markets.

4. Urban Policy Collaborations

To strengthen the real-world applicability of future housing market analyses, we recommend engaging in interdisciplinary collaboration with professionals from various socioeconomic fields, such as urban planning, real estate, and public policies etc. The professionals from these fields can provide insights and actionable strategies for urban and housing development, real estate investment, and renovations. Their involvement can help interpret quantitative findings into actionable urban strategies, particularly when addressing issues such as housing affordability, infrastructure development, or policy-driven displacement.

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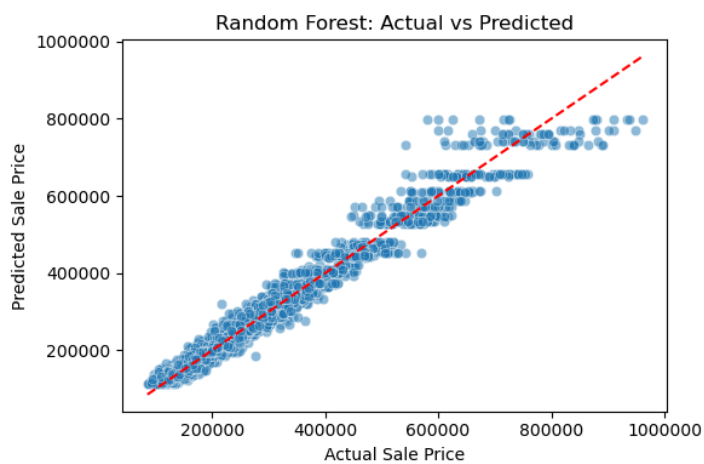
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%20value%20is%20estimated%20using%20smoothed%20and%20seasonally%20adjusted.rate%20estimates%20vary%20by%20region.

Appendix

This plot shows the predicted sale prices versus the actual sale prices for the Random Forest model trained on the original sales.parquet dataset.



A.

```

Top drivers of PC-1:
unemp_mean          0.384987
pop_avg             0.351127
unemp_trend         -0.349430
log_pop_avg         0.333789
pop_pct_change      -0.308746
pop_change          -0.275835
Stringency_Mean     0.268760
log_rent            0.249452
housing_unit_pct_growth -0.235240
pop_density         0.232068
housing_unit_density 0.217746
housing_unit_growth  0.180289
Region Type         0.000000
Name: PC1, dtype: float64

```

```

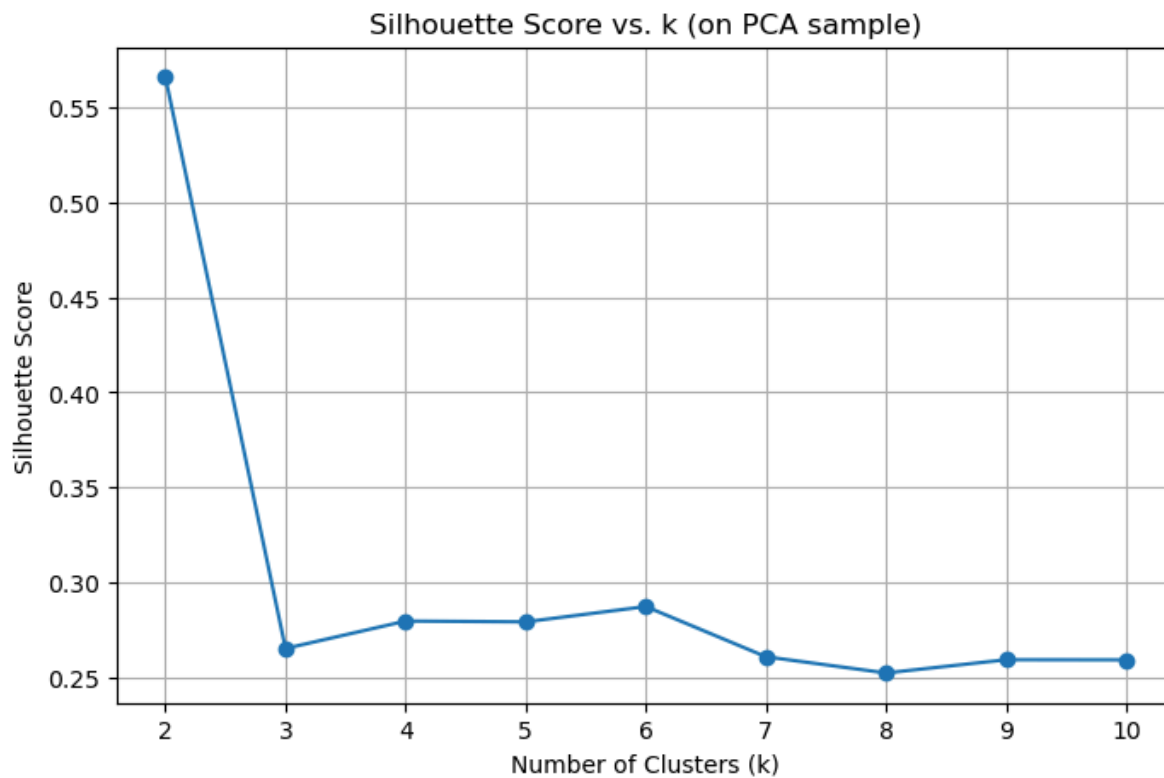
Top drivers of PC-2:
housing_unit_growth -0.510929
housing_unit_density 0.389980
pop_density         0.383248
pop_avg             -0.367162
log_pop_avg         -0.331921
housing_unit_pct_growth -0.263455
Stringency_Mean     0.255525
pop_pct_change      -0.176703
unemp_trend         0.118825
log_rent            0.087424
unemp_mean          -0.061881
pop_change          -0.059097
Region Type         -0.000000
Name: PC2, dtype: float64

```

In the appendix, the top drivers of PCA components demonstrate which features contributed most to the reduced dimensional structure:

- PC-1 is driven by unemployment rates, population change, stringency, highlighting pandemic-related socioeconomic disruption
- PC-2 has more focus on housing supply growth and density, demonstrating structural housing market differences.

B.



The silhouette score graph indicates that $k=3$ was the optimal number of clusters for KMeans.

The sharp drop off after 3 confirms this grouping yields the best internal cohesion, validating the selection of three regional profiles.

Source Code

Home Values EDA

 DAT490CapstoneHV.pdf

New Constructions EDA

 DAT490CapstoneNC.pdf

Time on Market EDA

 Data490_Time_On_Market.pdf

Rental Dataset - Munging/Cleaning

 Rentals_DS_Cleaning_Merging.pdf

Rentals Dataset - Feature Engineering/Modeling

 DAT_490_Capstone_rentals_FE_model.pdf

Covid Stringency Index Dataset

 Covid_stringency_index.pdf

Population Dataset with maps

 Curr_Population_map.pdf

Housing RQ1 PDF

 RQ1Code (1).pdf

